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LKS Faculty of Medicine
**Department of Orthopaedics
& Traumatology**
香港大學矯形及創傷外科學系

Osteoporotic Related Fractures

Professor Christian Fang
Clinical Associate Professor

Objectives



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- Osteoporotic Hip fractures
 - How to diagnose
 - How to treat
 - Pathological considerations
- Osteoporosis
- Other fragility fractures
 - Distal radius
 - Vertebral collapse

Patient 1



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- 82 years old woman
- Walk with stick
- IHD on aspirin
- S/F on level ground
- Left hip pain
- Unable to walk



Physical exam



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- Left hip painful to any movement
- Lower limb shortened
- Externally rotated





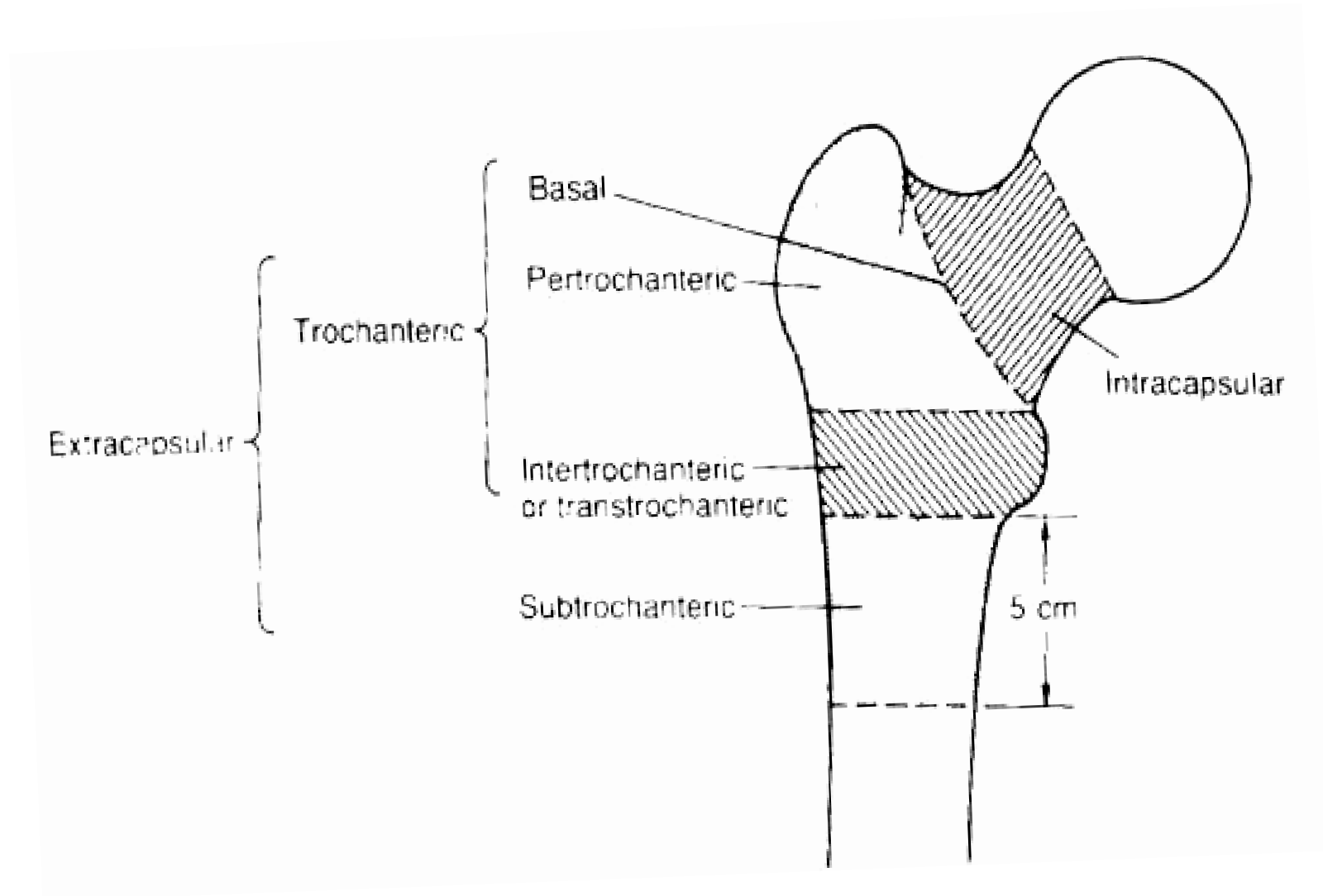
Shenton's line?



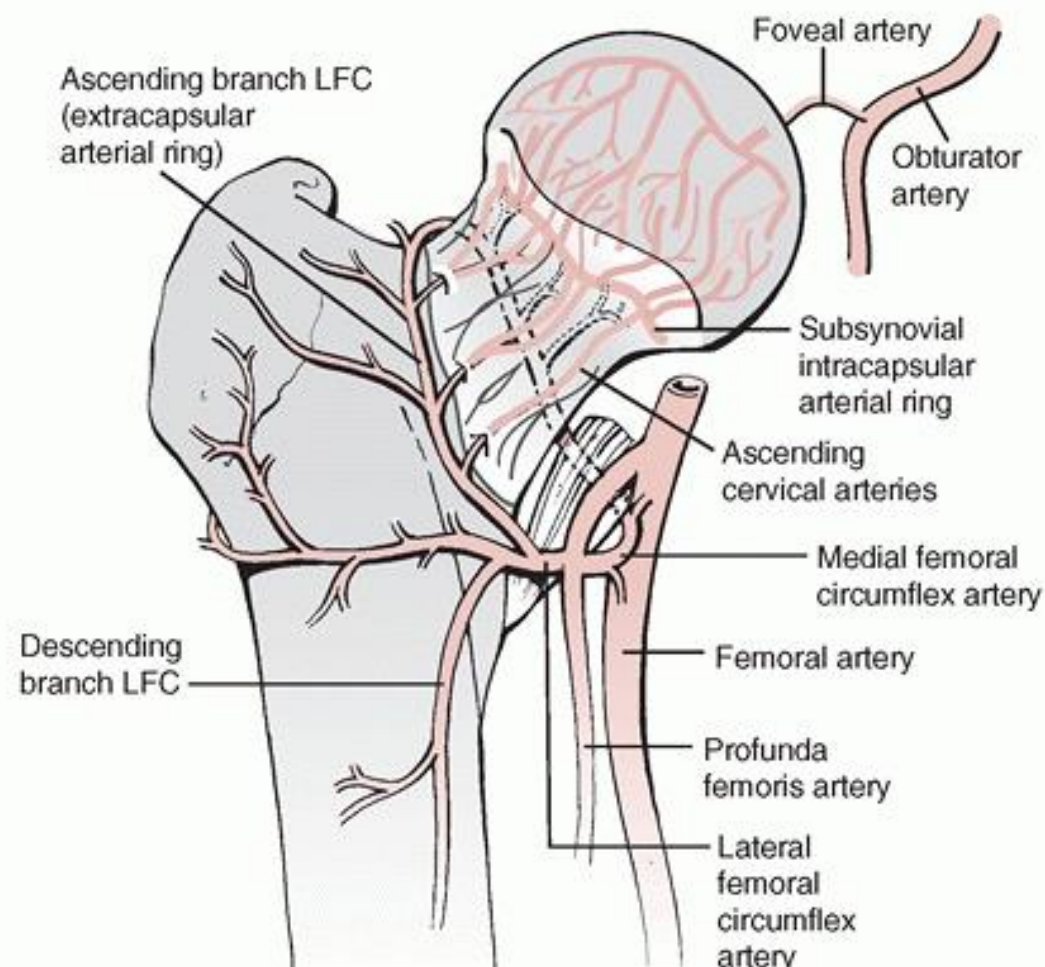
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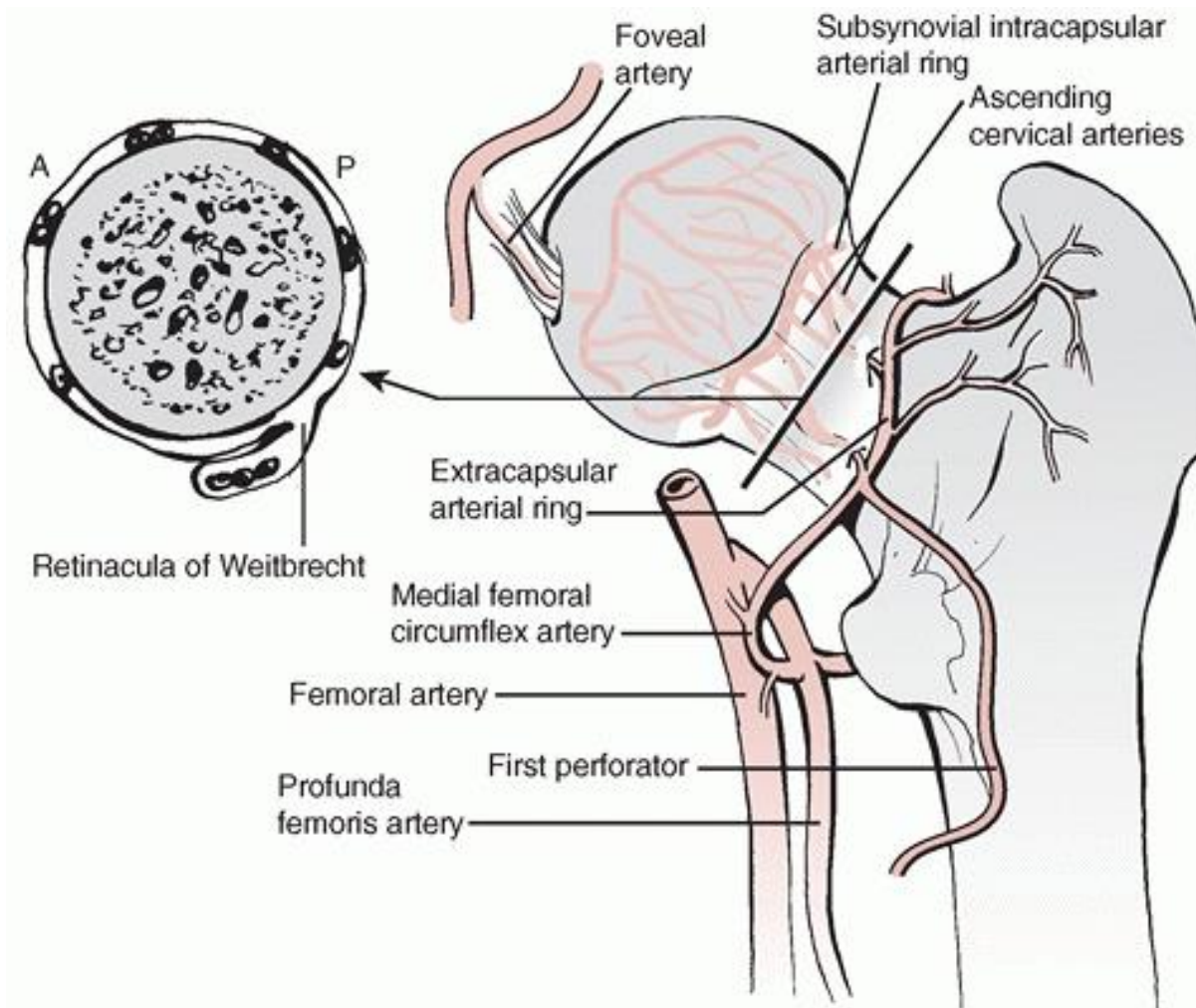
Shenton's Line

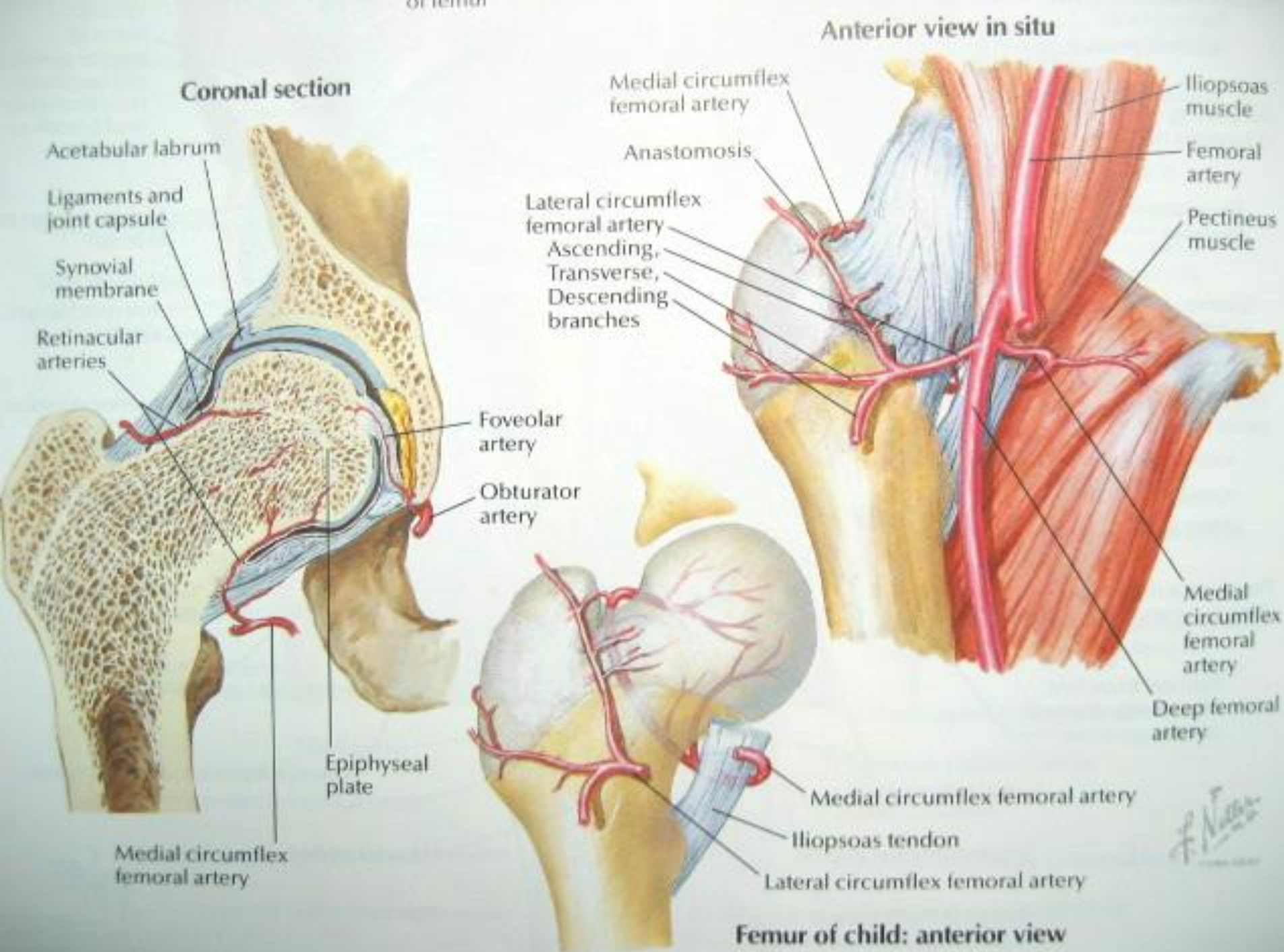


Frontal view



Posterior view





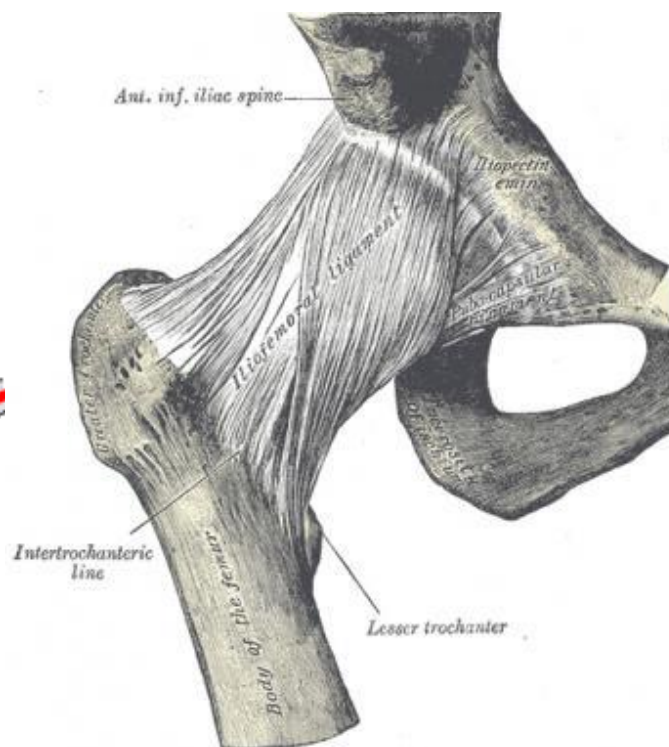
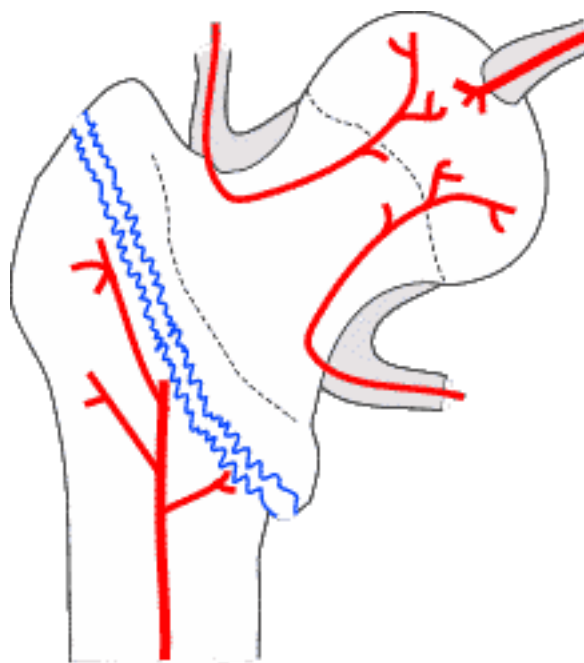
Blood supply to femoral head



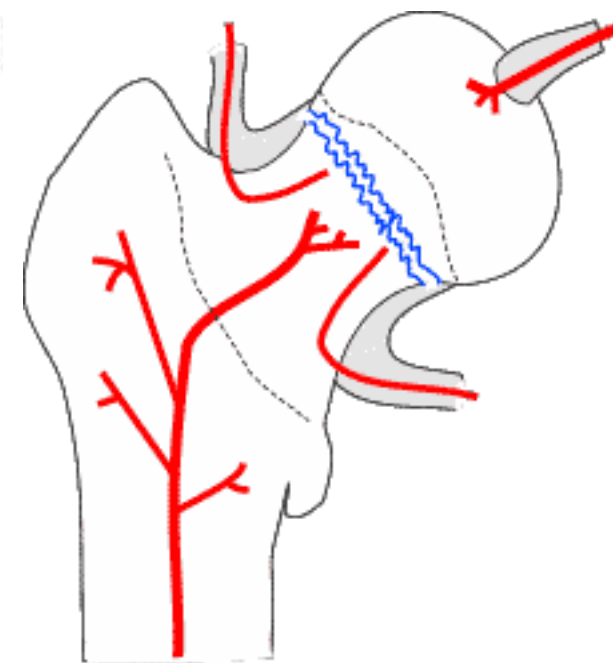
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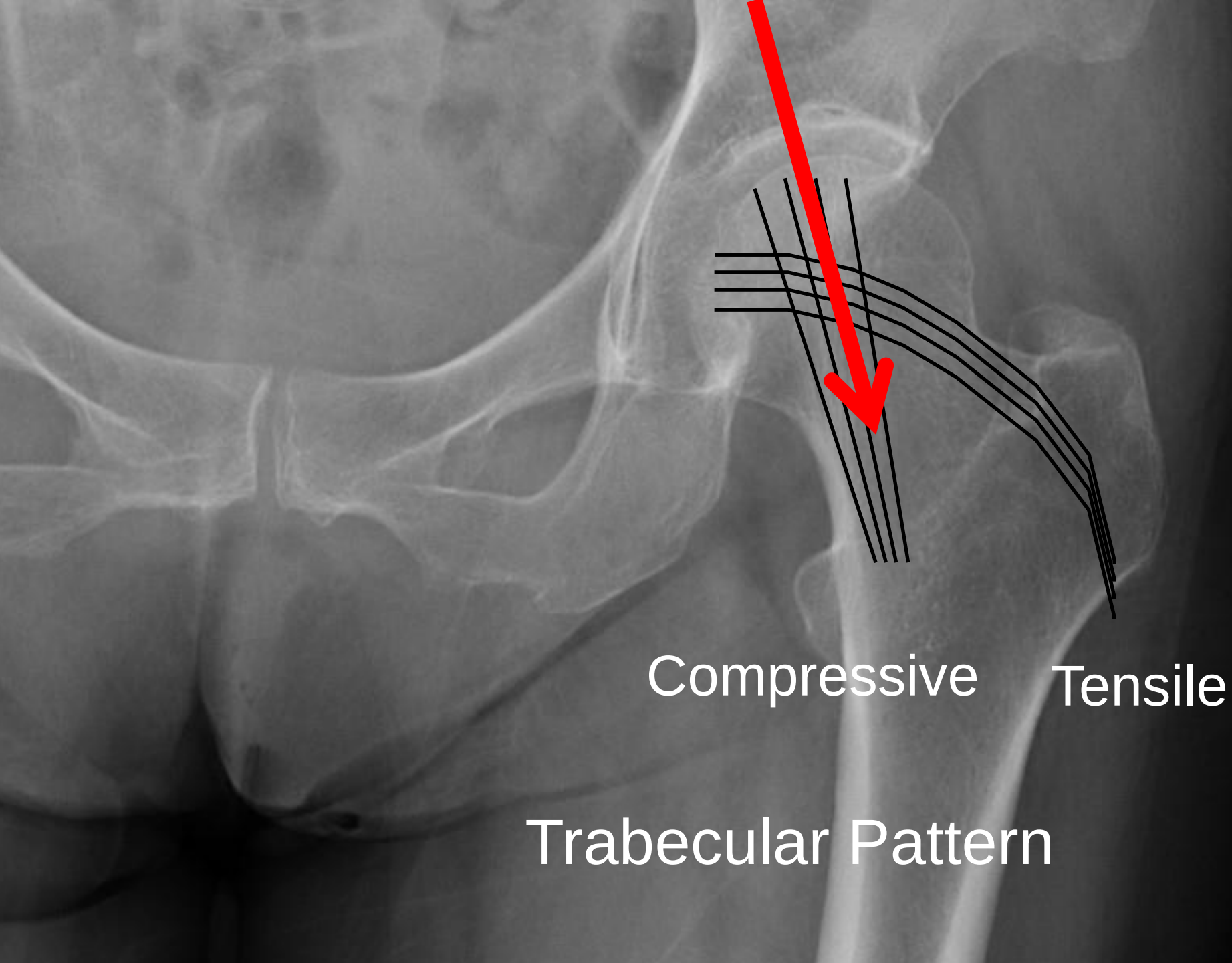
- Retrograde flow
- Reticular artery from medial and lateral circumflex arteries
- Ligamentum Teres (not effective in adults)
- Epiphyseal blood supply

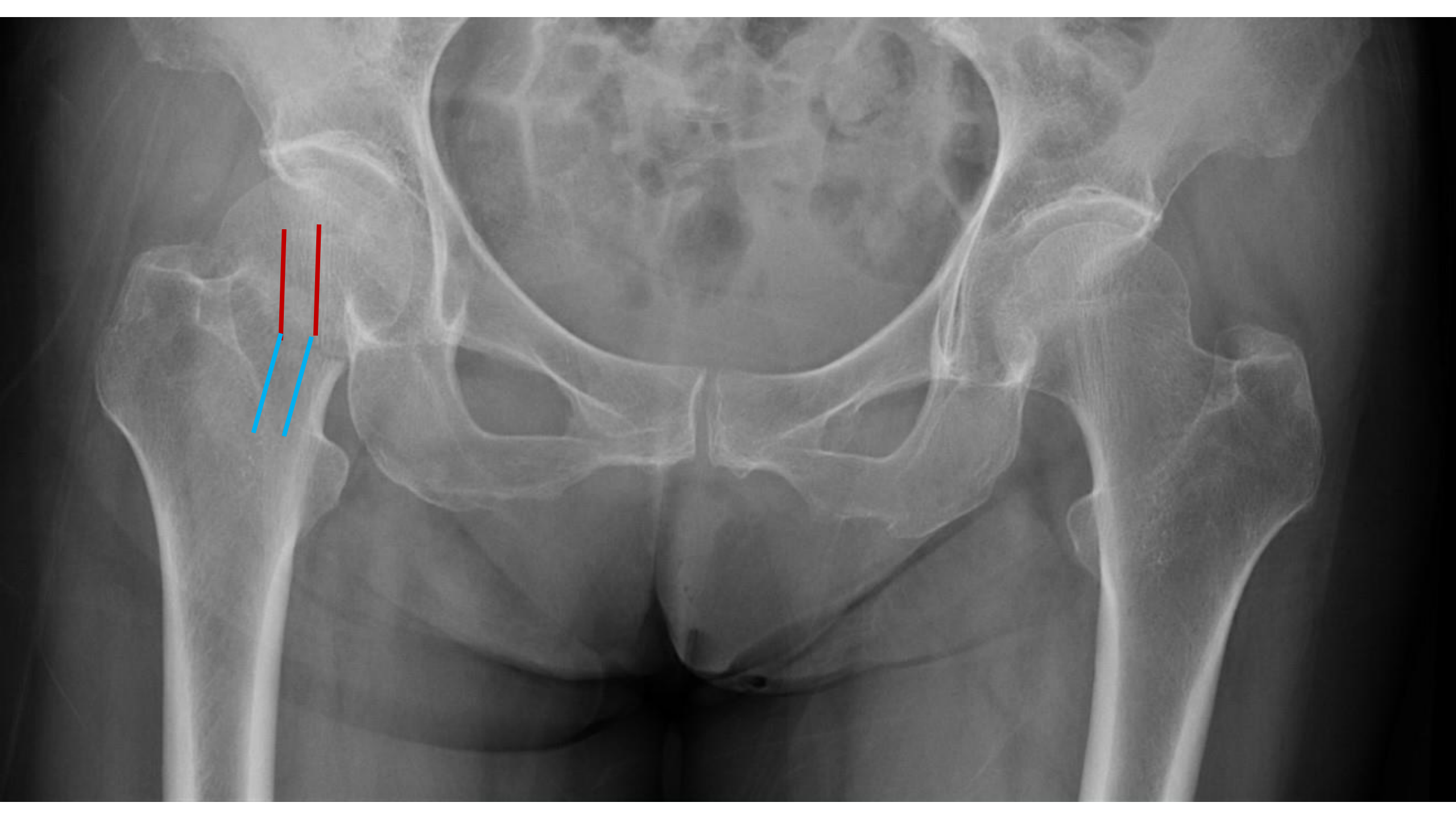
Extra-capsular
fracture



Intra-capsular
fracture



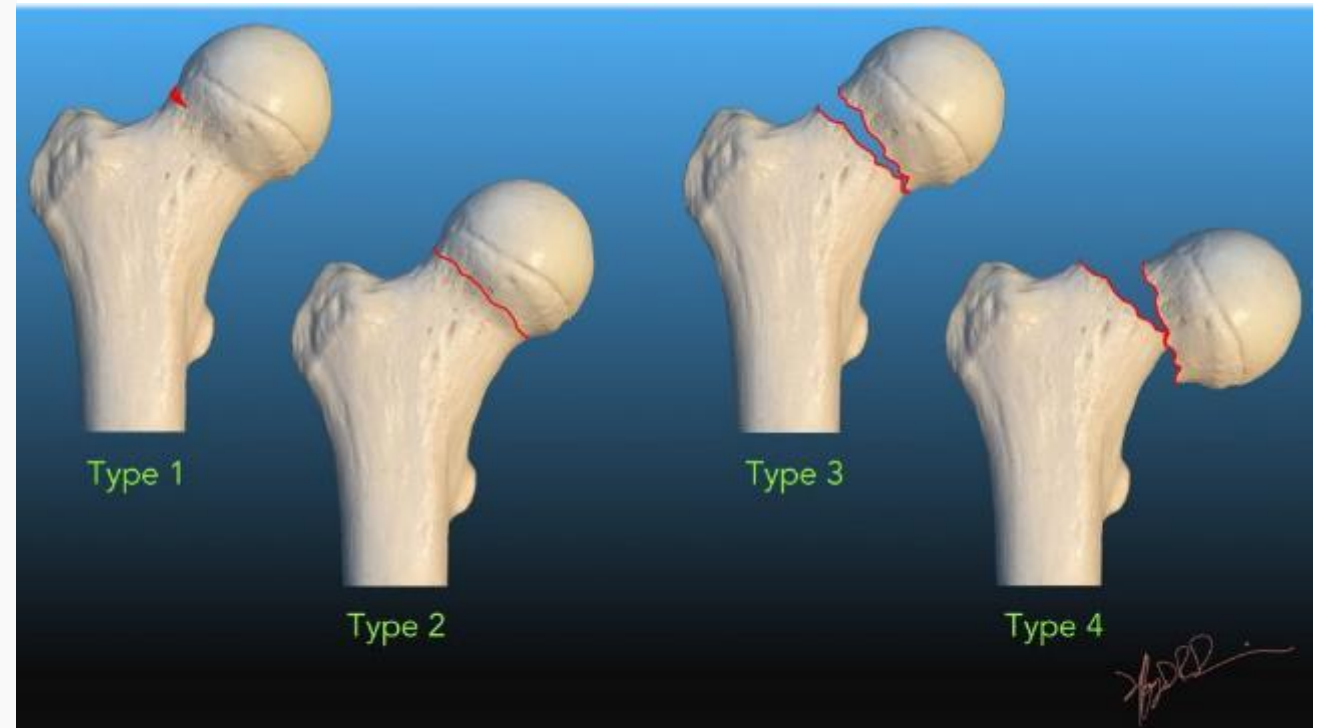
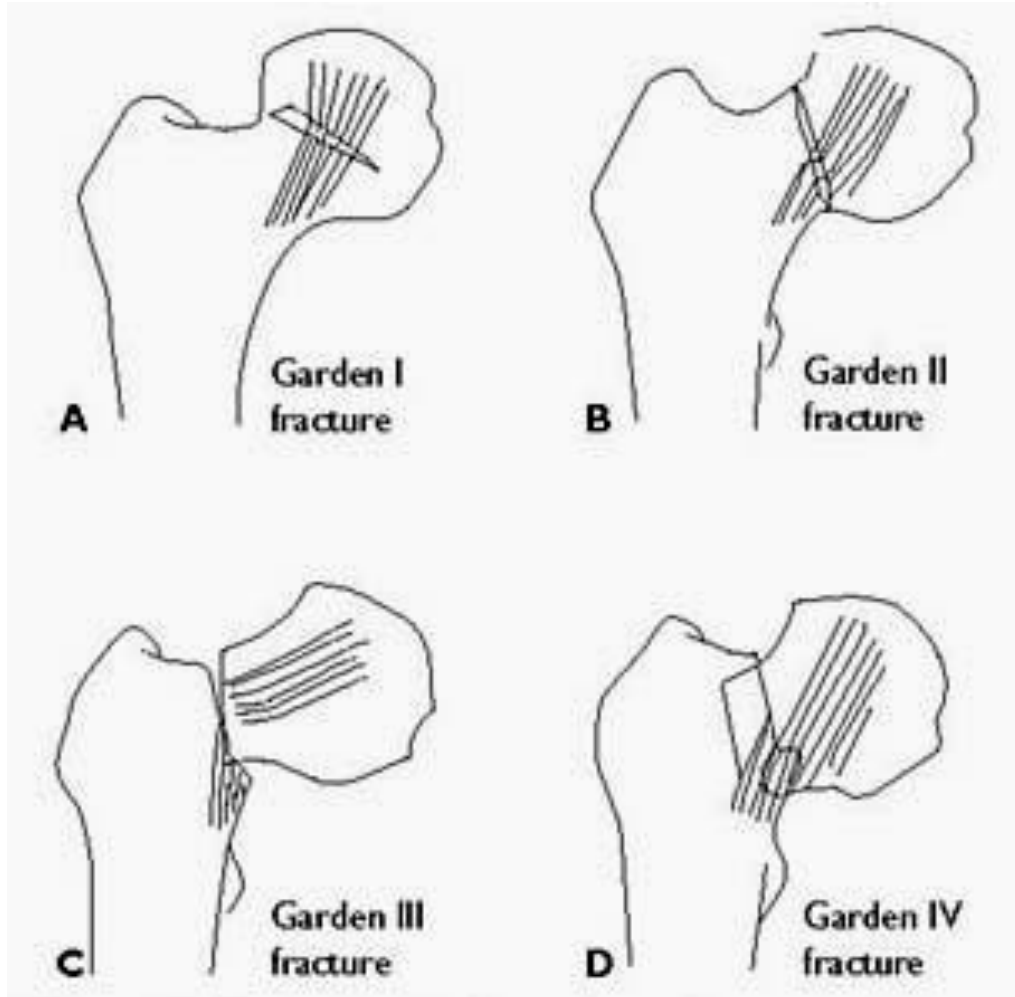




Garden classification



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Fracture NOF



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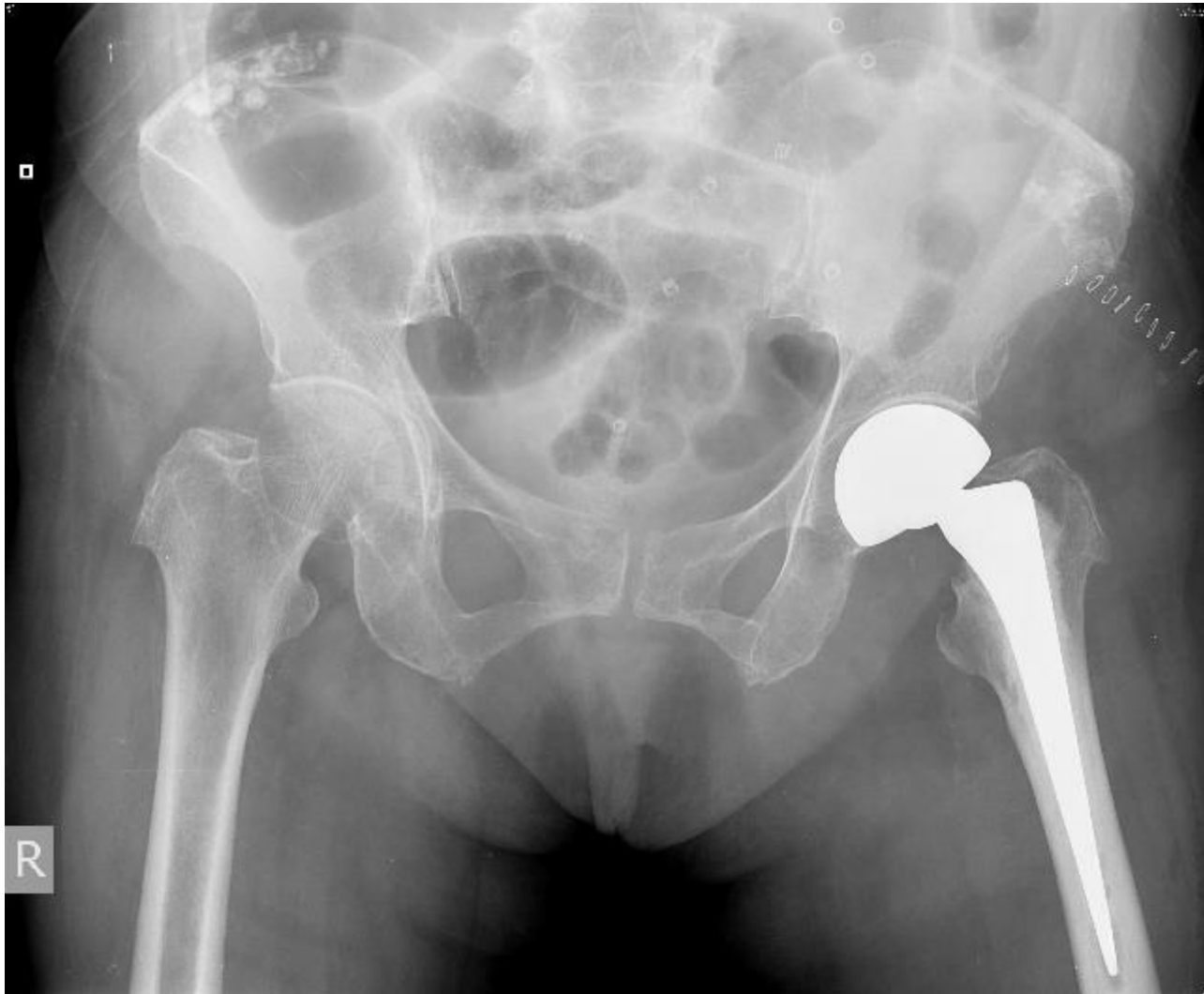
- Risk of AVN and non-union
 - Minimally displaced #-----<10%
 - Displaced fracture----→50%



Tx: Hemiarthroplasty



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Hemiarthroplasty



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**monoblock
Austin Moore
HemiArthroplasty(AMA)**



**Modular
Cemented Unipolar
Hemiarthroplasty**



**Modular
Cemented Bipolar
Hemiarthroplasty**





Mono-Bloc Prosthesis

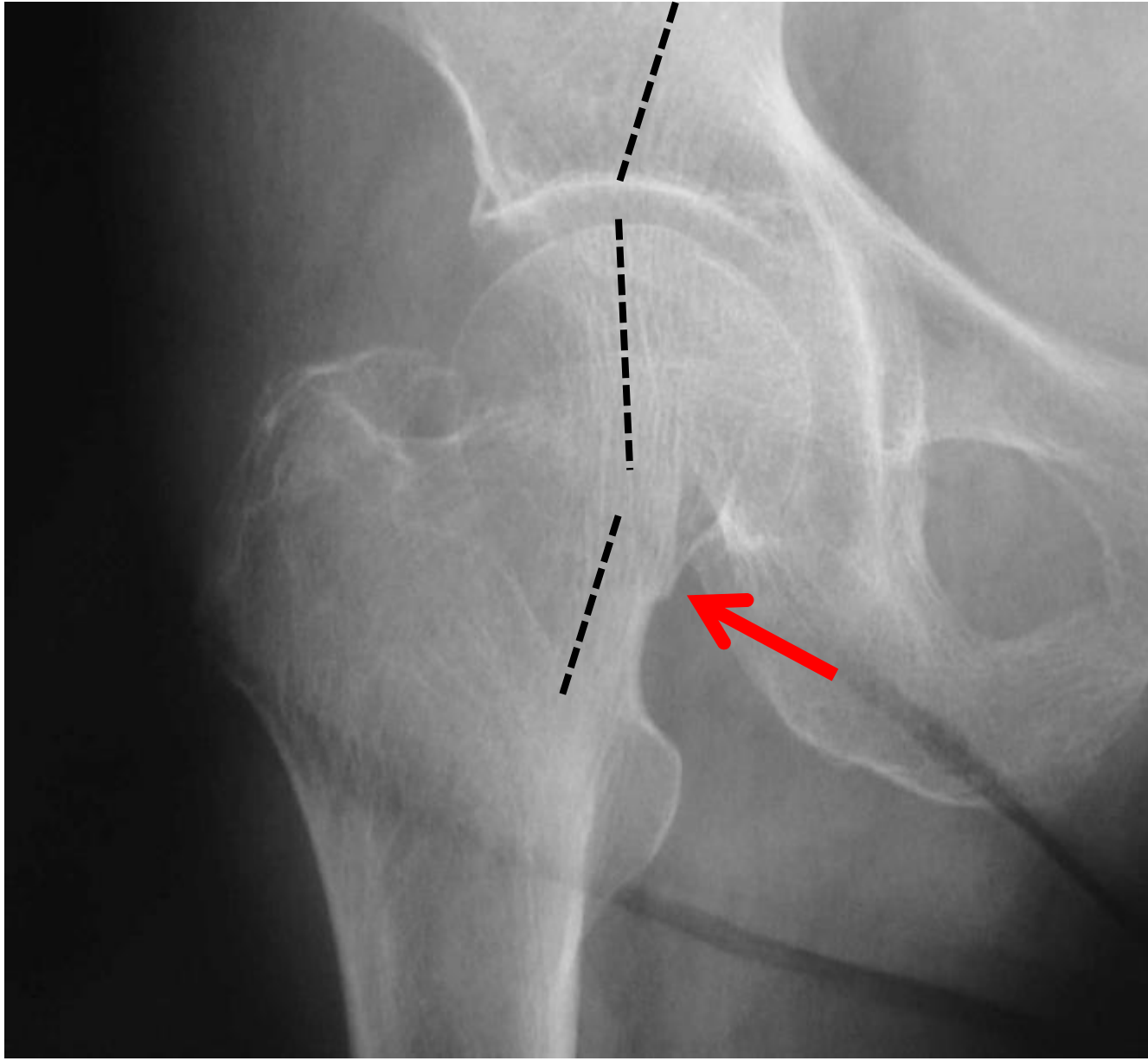


Modular Prosthesis

Valgus Impacted fracture (Garden 1)



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Risk factors



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- *Increased risk of falls*
 - Concurrent medical illness
 - Drugs- tranquillisers, alcohol
 - Dementia / Sarcopenia / Visual impairment
- *Reduction in protective responses*
- *Loss of local shock absorbers*
- *Loss of bone strength*
 - Drugs- corticosteroids, anticonvulsants, thyroxine, alcohol
 - Smoking
 - Vit D deficiency
 - Calcium deficiency
 - Physical inactivity

Fracture NOF considerations



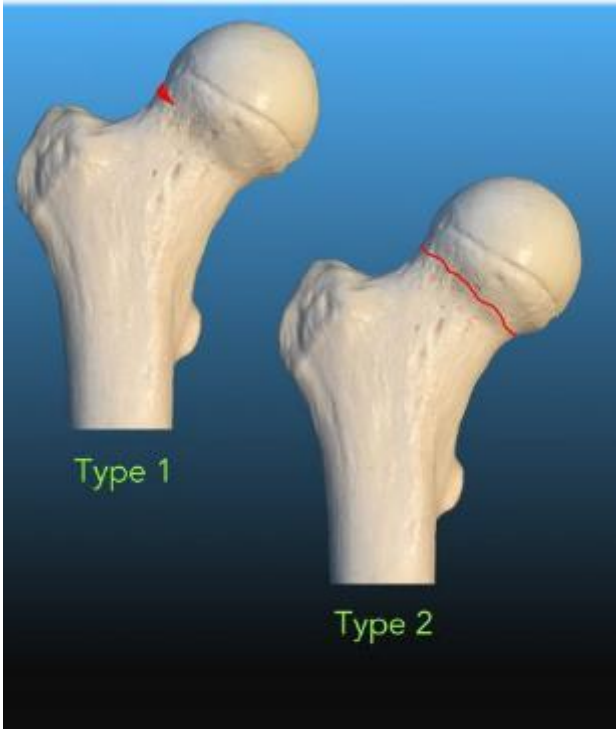
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- AGE
- Displacement
- Save the femoral head in young age / undisplaced
- Other factors to consider:
 - Time of presentation-----AVN (6hr)
 - Patient's general health
 - Risk factors for internal fixation
 - ---Osteoporosis, too comminuted
 - ---Sepsis

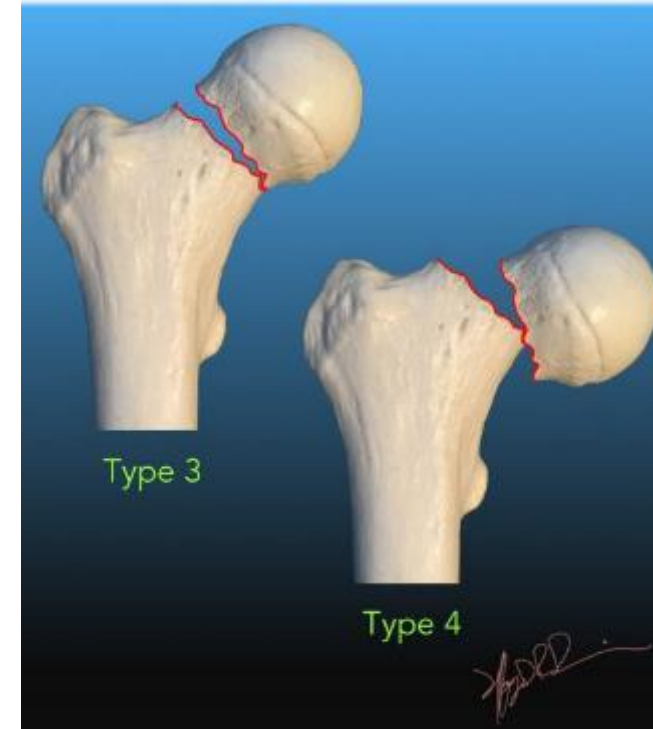
Garden classification



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Undisplaced
→ Internal fixation to prevent displacement



Displaced
Young age → Reduction and Internal fixation
To salvage joint

Old age → Partial hip replacement
(Hemiarthroplasty)
To prevent reoperation and control pain

Factors with increase risk for fixation failure



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- Old age
- Osteoporosis
- Vertical fracture pattern
- Delayed presentation



Case 2

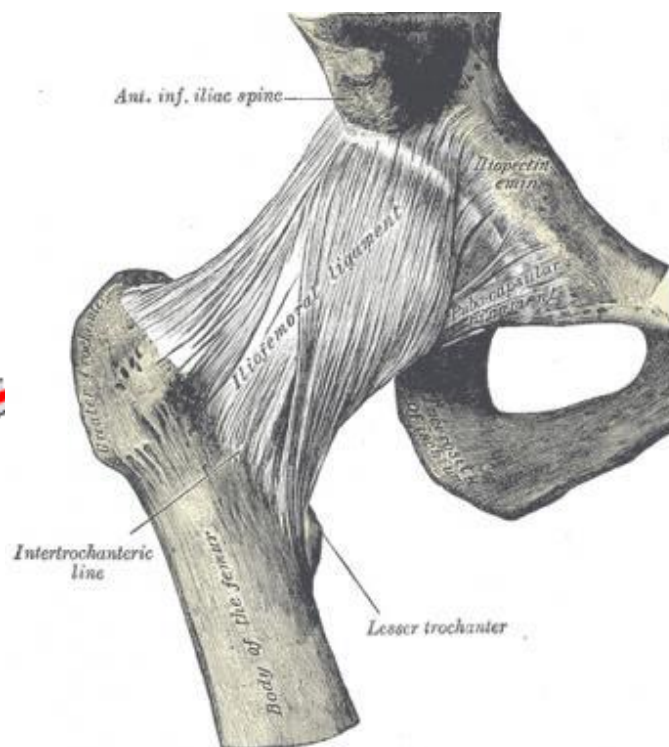
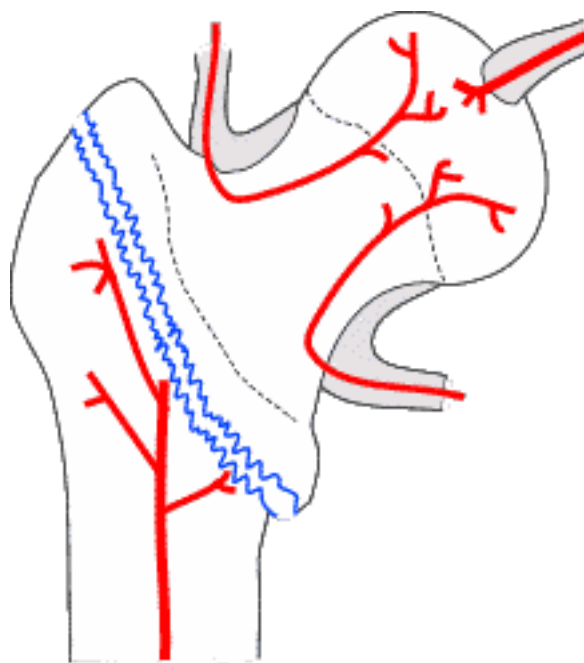


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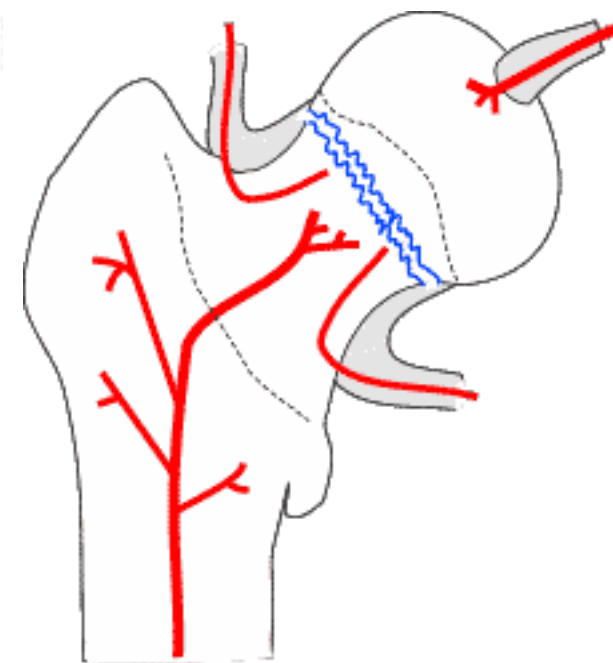
- 85 years old woman
- Fall on level ground
- Right hip pain



Extracapsular
fracture



Intracapsular
fracture

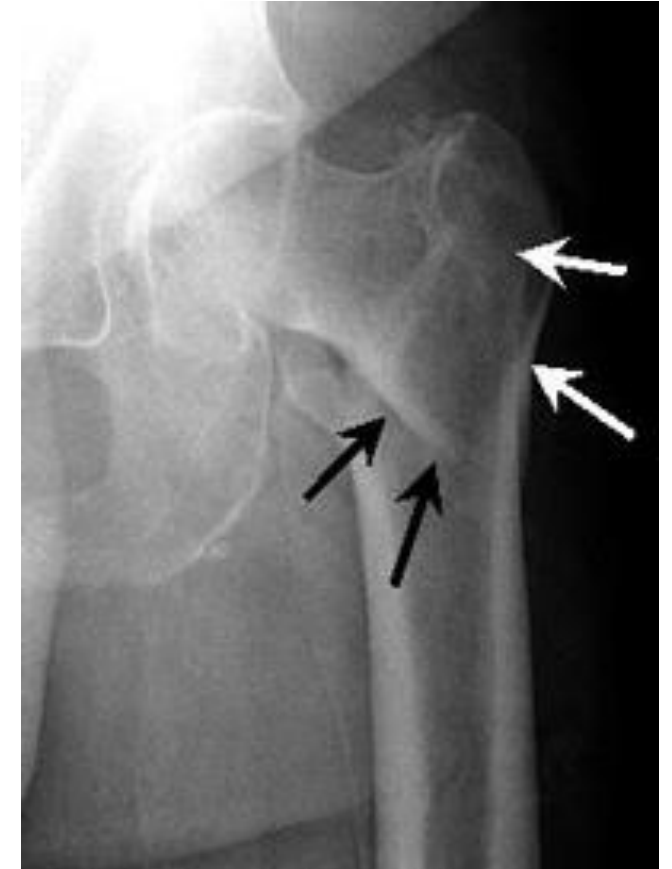


Reduction & Internal Fixation



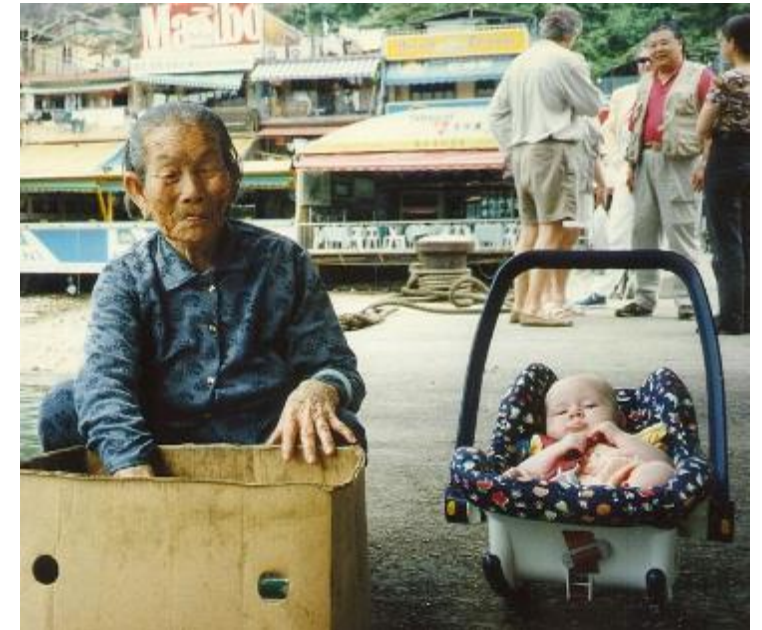
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- Restoration of neck-shaft angle
 - Avoid varus alignment of proximal fragment
- Stabilize fracture to facilitate early mobilization



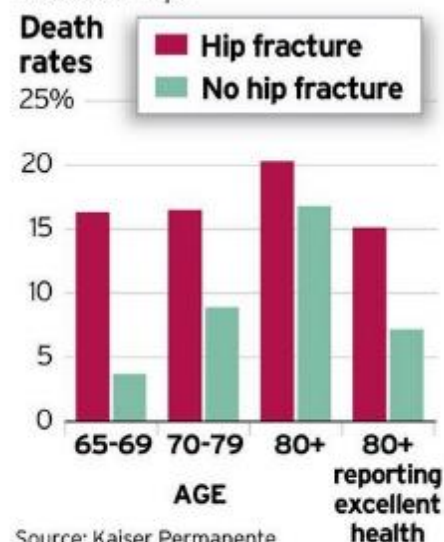
Aim of treatment

– Return to pre-injury status

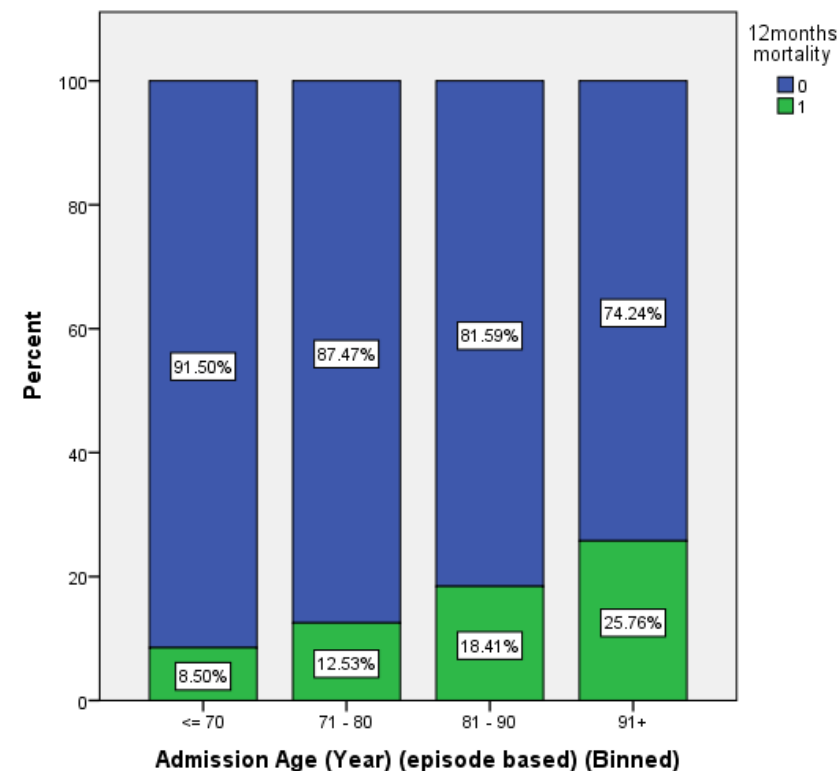


Hip fracture and mortality

Death rates within a year after a hip fracture are up to 4.5 times higher than death rates among same-age women who have not broken a hip.



LeBlanc et al Arch Intern Med. 2011



HKSAR PRC



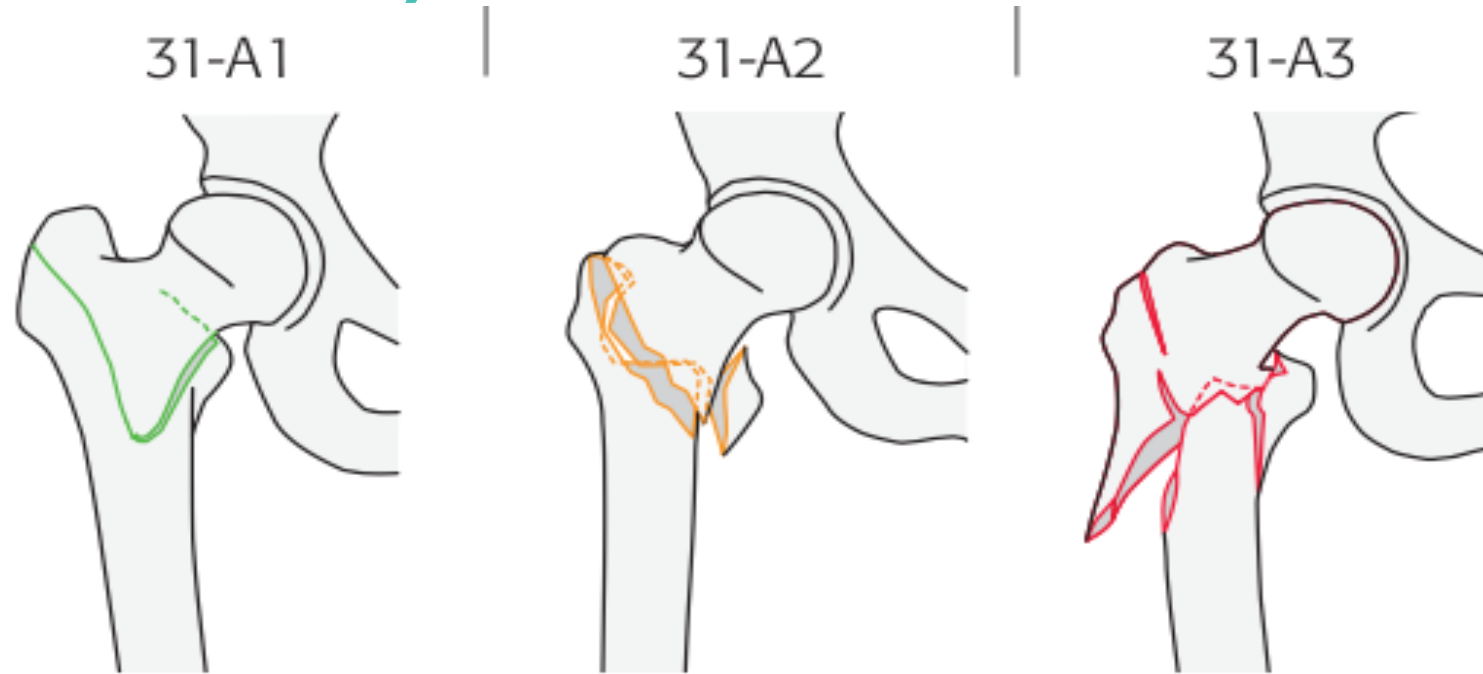
Appropriate implant



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- Recognize the unstable patterns and choose an intramedullary device to reduce risk of fixation failure

Intertrochanteric fracture (AO classification)



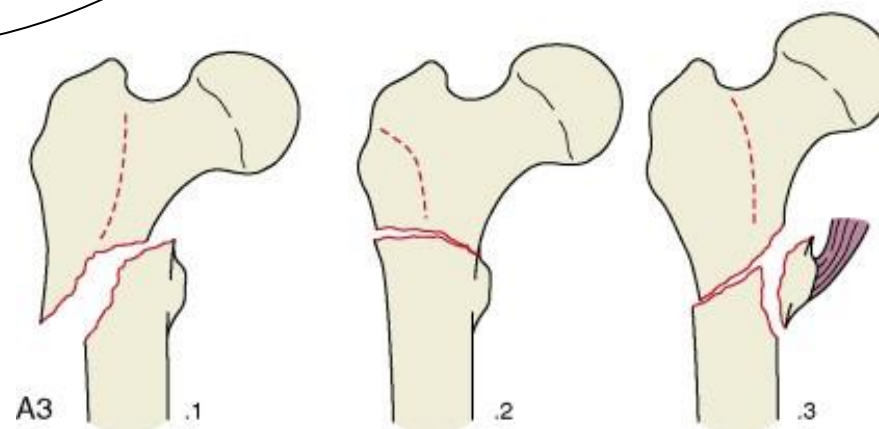
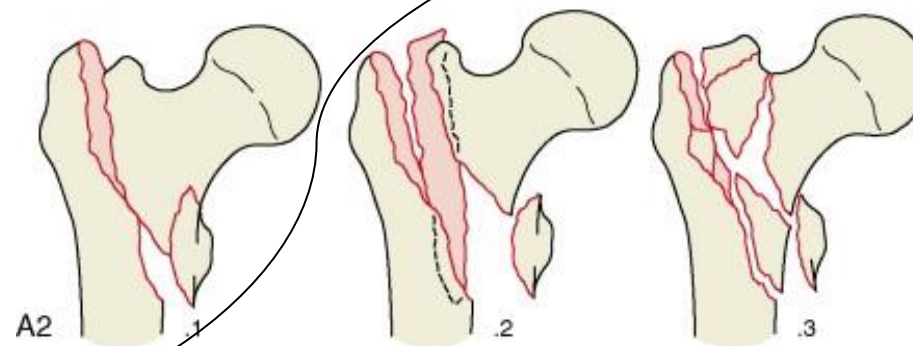
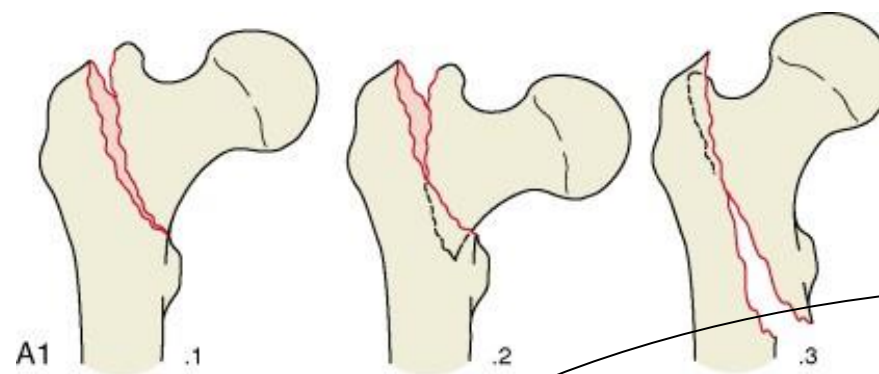
31-A extraarticular fracture, trochanteric area

31-A1 pertrochanteric simple

31-A2 pertrochanteric multifragmentary

31-A3 intertrochanteric

Stable



Unstable

2 Types of per-trochanteric fractures

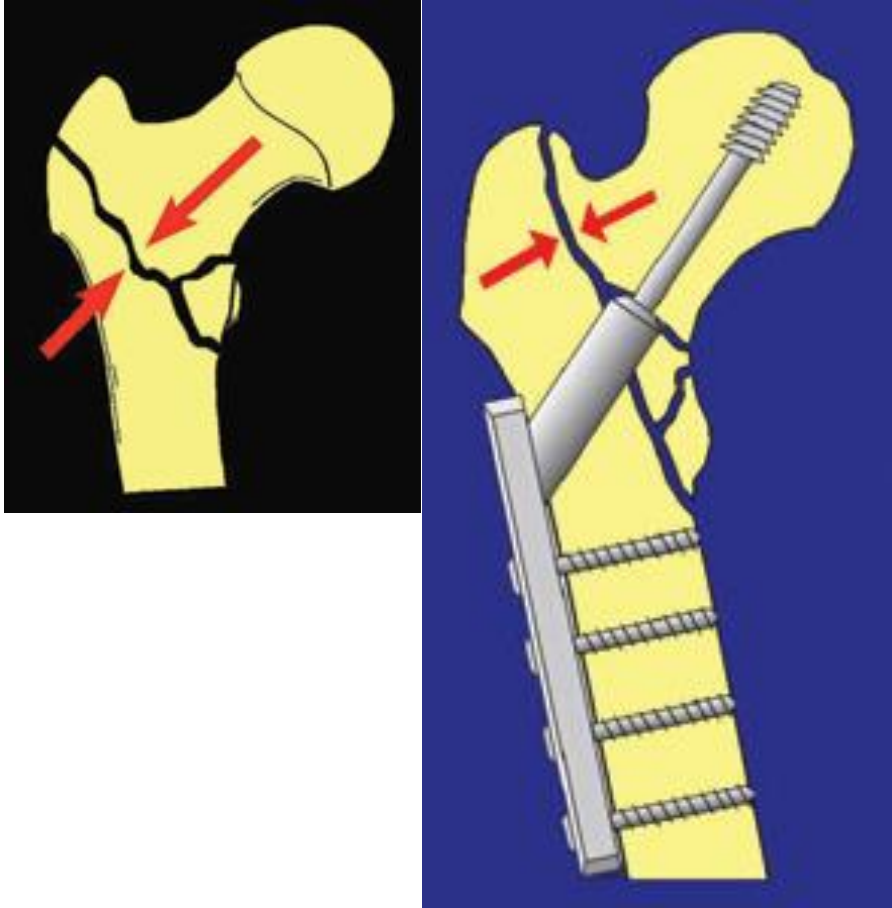


- Stable
 - Oblique pattern
 - Minimal comminution
 - Repair with Sliding hip screw
- Unstable
 - Reverse obliquity
 - Subtrochanteric extension
 - Comminution
 - Repair with intramedullary device (cephalomedullary nail)

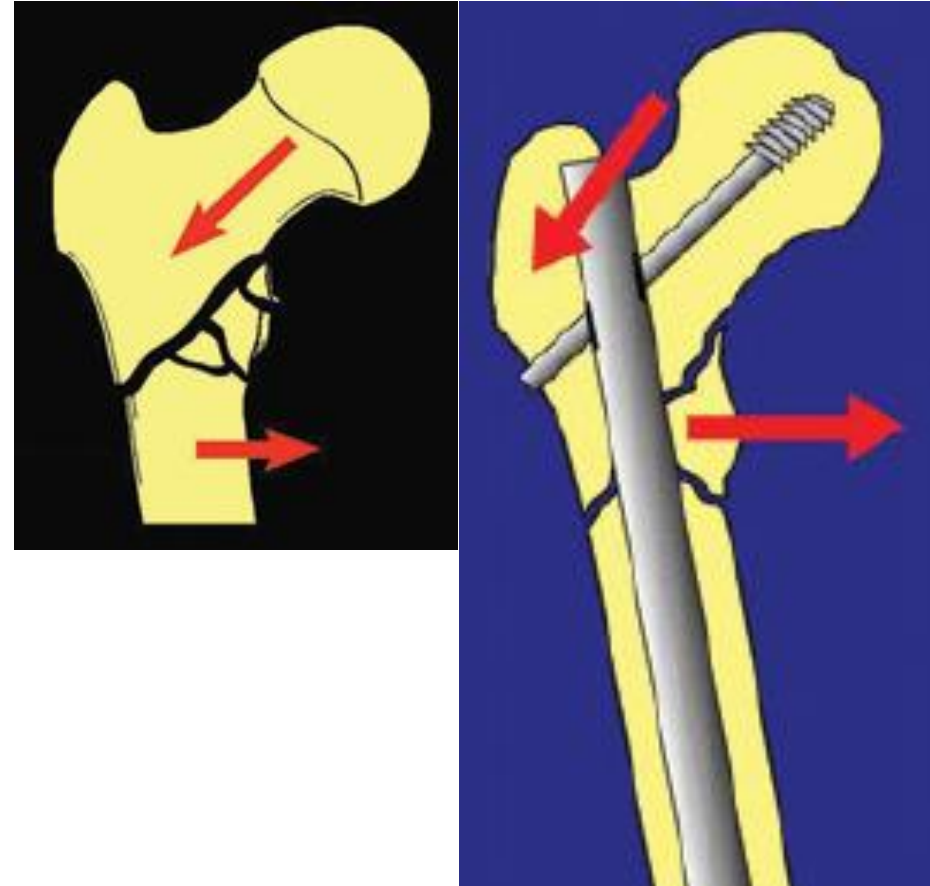
Stable vs Unstable



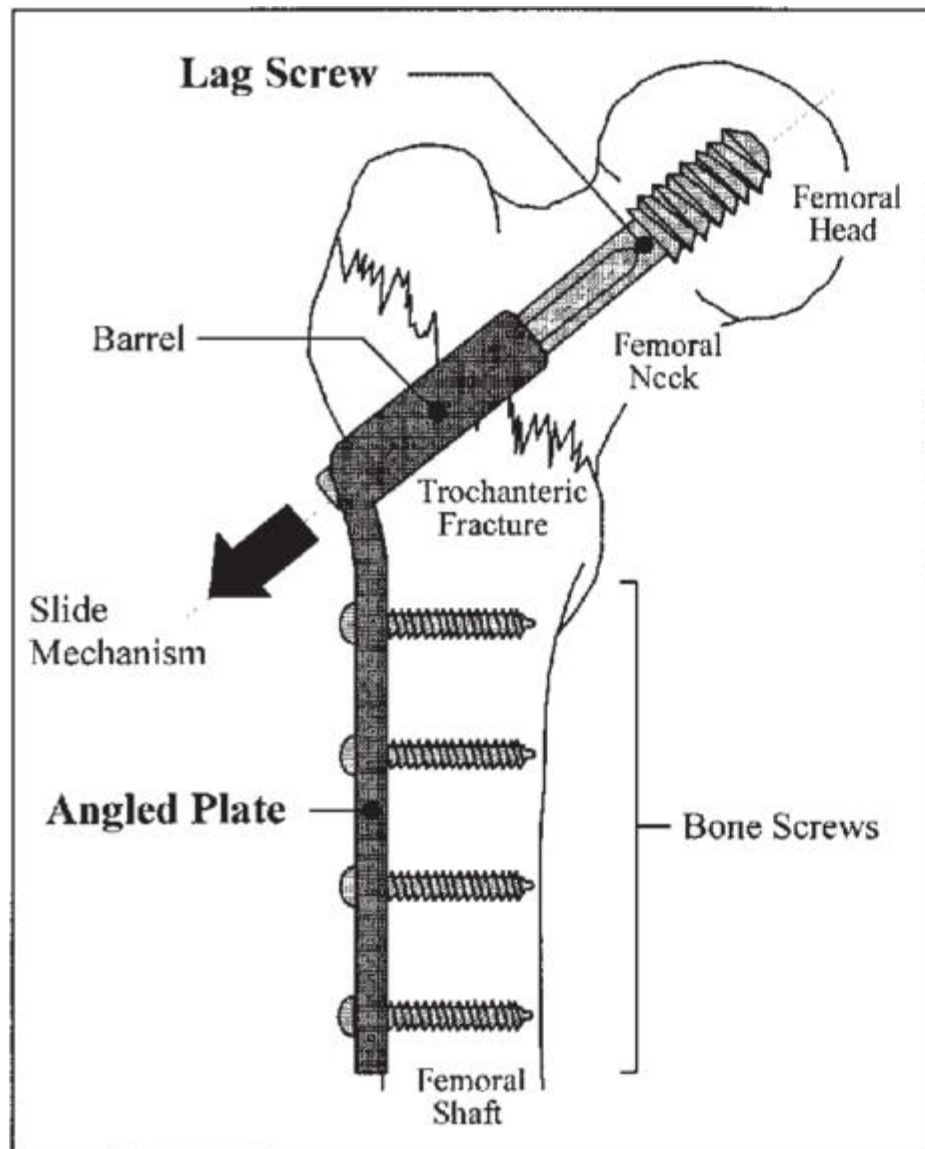
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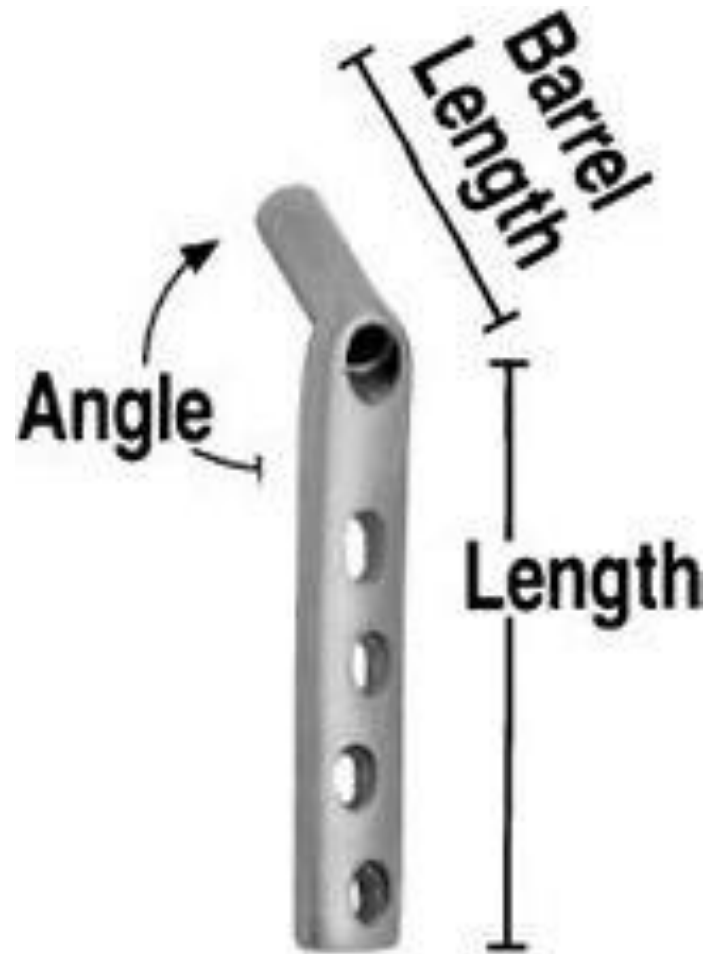


Sliding hip screw



Cephalomedullary nail



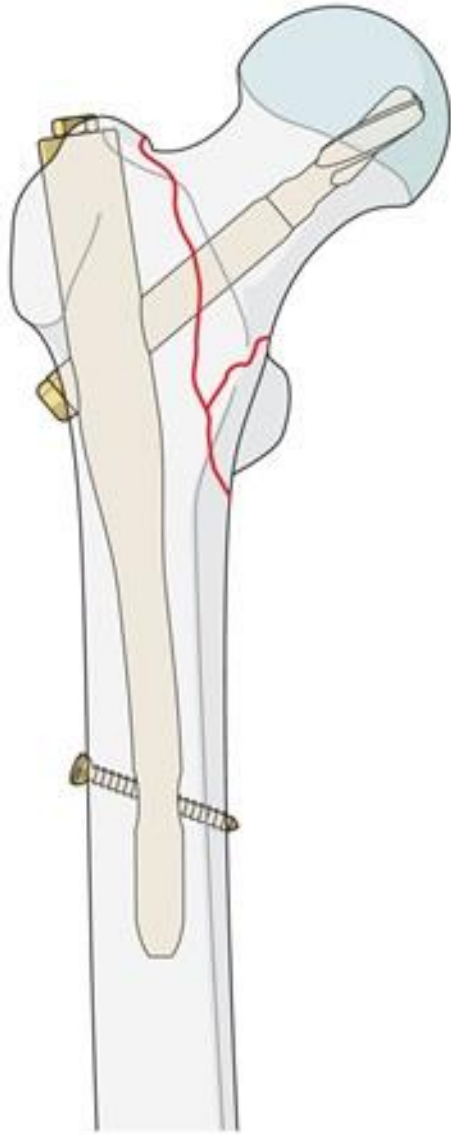


- Length
 - 2 holes as good as 4 holes
 - Bolhofner JOT 1999
 - McLoughlin JOT 2000

Cephalomedullary Nail



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Differences



- Sliding hip screw
 - Cheap ~\$1000
 - Technically easier
 - Less risk of iatrogenic fracture
 - Requires more simple fracture patterns
- Cephalomedullary nail
 - Expensive ~\$7000
 - Technically difficult
 - Intramedullary fixation is mechanically stronger
 - Less change for excessive collapse

Case 1 – other common



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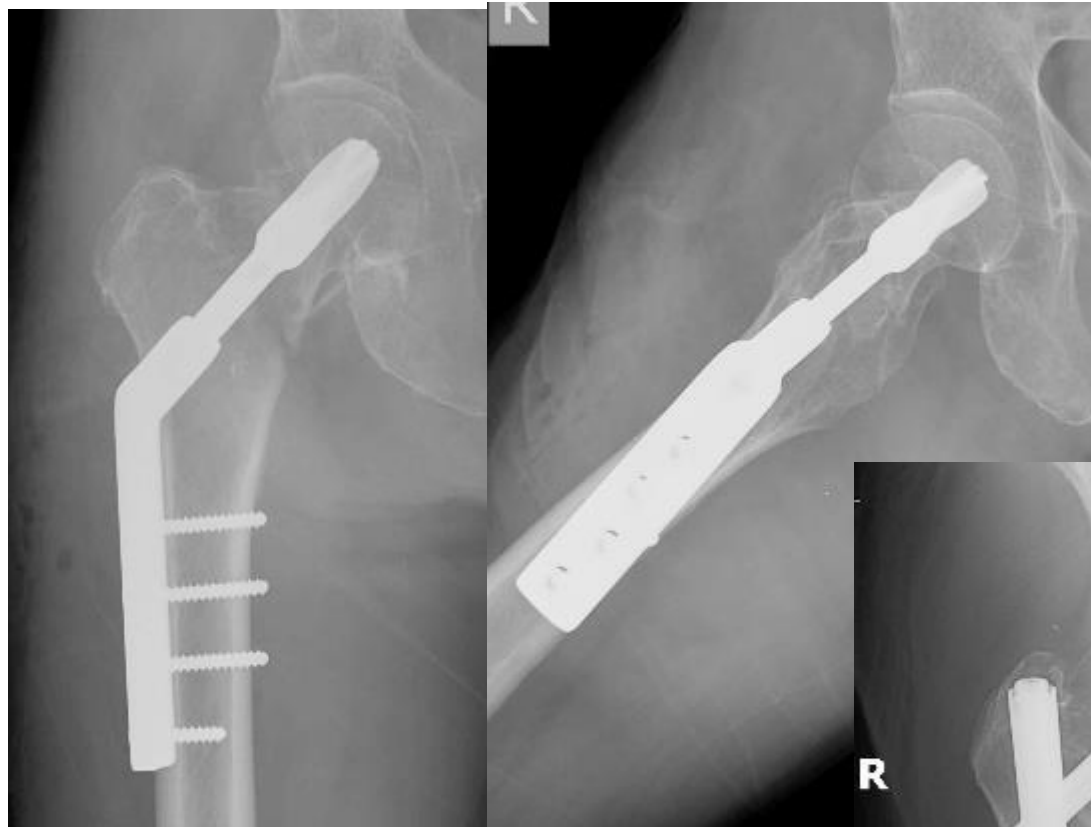


Case 1 – other common



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Cephalomedullary nail

Dynamic hip screw



Conclusion



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- Intracapsular neck of femur #
- Worry of AVN, nonunion
 - Young patient – reduction and internal fixation
 - Old patient - hemiarthroplasty if displaced, internal fixation if undisplaced
- Trochanteric fracture
 - Mechanics
 - Cephalomedullary device for unstable pattern

Osteoporosis



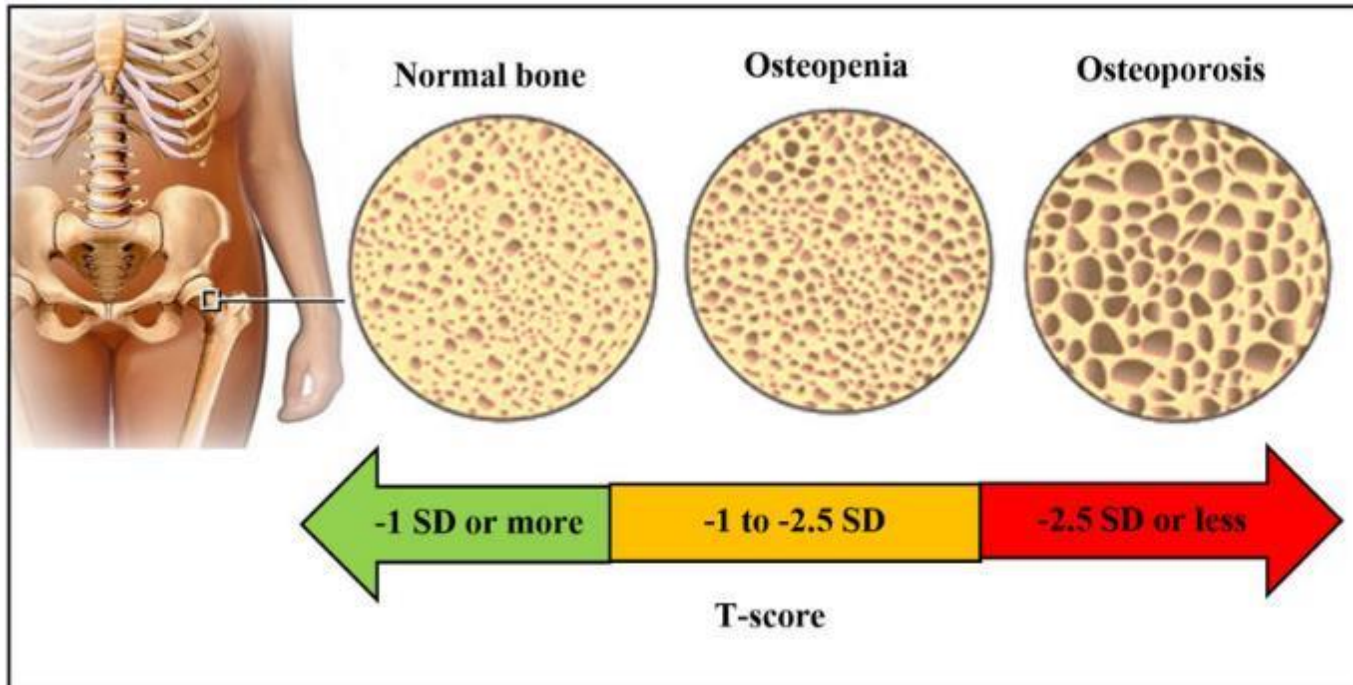
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- Definition
 - Systemic skeletal disorder
 - Characterized by low bone mass,
 - Micro-architectural deterioration of bone tissue
 - Leading to bone fragility
 - Consequent increase in fracture risk.

Osteoporosis diagnosis



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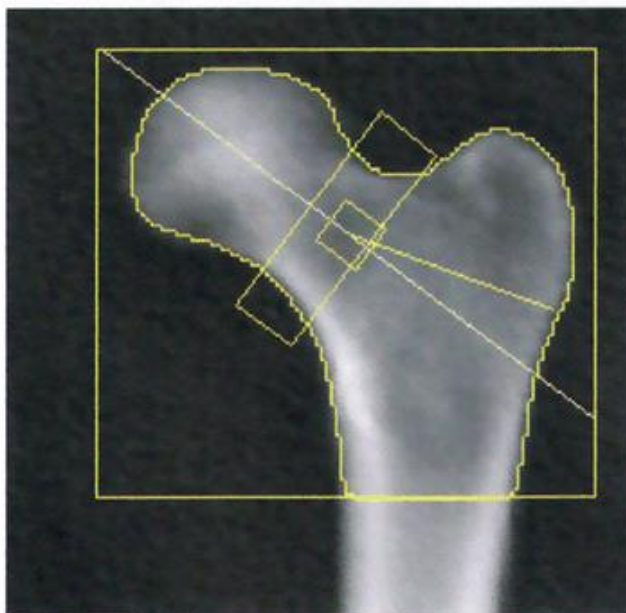


Image not for diagnostic use

k = 1.162, d0 = 61.6

101 x 91

NECK: -49 x 15

Scan Information:

Scan Date: 23 May 2013

ID: K0523131R

Scan Type: a Left Hip

Analysis: 23 May 2013 17:39 Version 12.6:5
Left Hip

Operator:

Model: QDR 4500C (S/N 47998)

Comment:

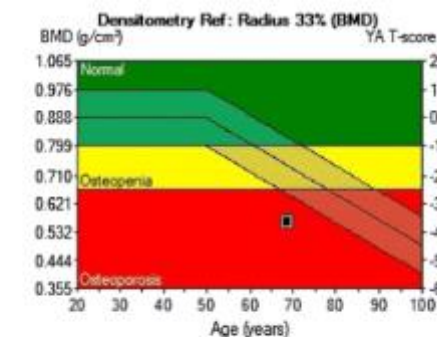
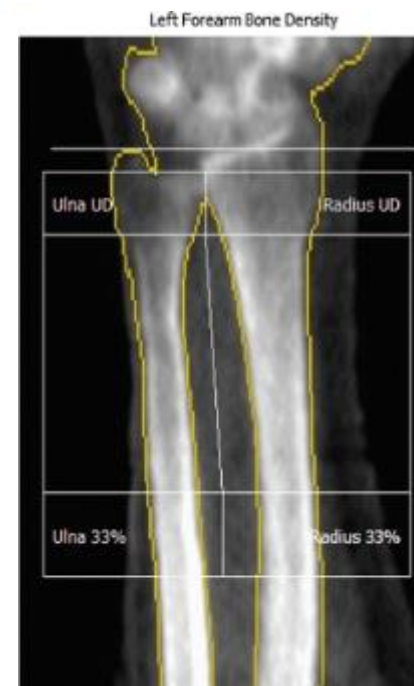
DXA Results Summary:

Region	Area (cm ²)	BMC (g)	BMD (g/cm ²)	T - score	PR (%)	Z - score	AM (%)
Neck	4.61	2.87	0.622	-2.0	73	-0.8	87
Troch	10.15	5.18	0.510	-1.9	73	-1.1	82
Inter	17.34	14.83	0.855	-1.6	78	-1.0	85
Total	32.11	22.88	0.712	-1.9	76	-1.0	85
Ward's	1.11	0.50	0.448	-2.4	61	-0.6	87

Total BMD CV 1.0%, ACF = 1.023, BCF = 1.011, TH = 1.136

WHO Classification: Osteopenia

Fracture Risk: Increased



Region	¹ BMD (g/cm ³)	² Young-Adult T-score	³ Age-Matched Z-score
Radius 33%	0.560	-3.7	-2.0

Name: Express Scans, 2
Patient ID:
DOB: August 24, 1944

Sex: Female
Ethnicity: White

Height: 65.0 in
Weight: 150.0 lb
Age: 61



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Referring Physician:

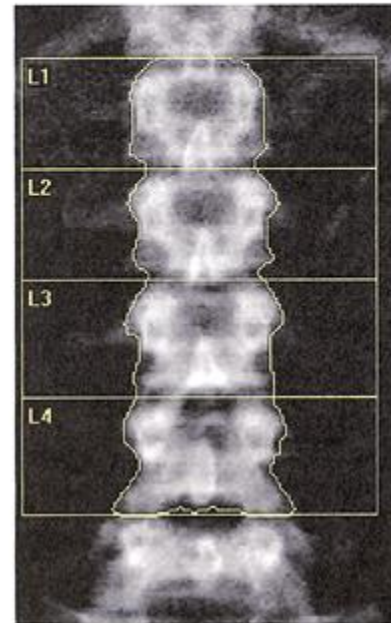


Image not for diagnostic use
k = 1.138, d0 = 48.0
116 x 149

Scan Information:

Scan Date: November 12, 2005 ID: A11120501
Scan Type: x Lumbar Spine
Analysis: November 12, 2005 09:48 Version 12.4:3
Lumbar Spine

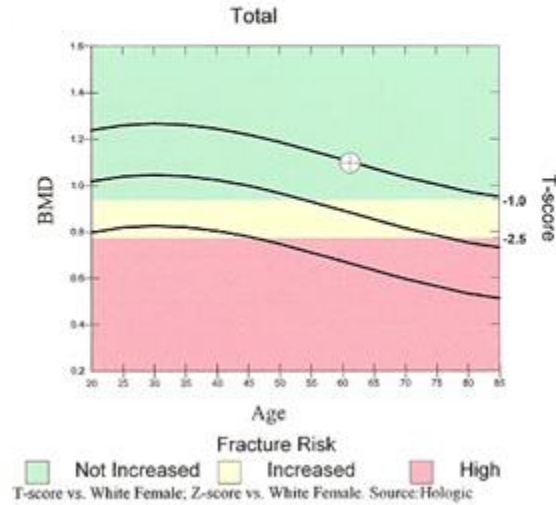
Operator:
Model: Discovery C (S/N 81202)
Comment:

DXA Results Summary:

Region	Area (cm ²)	BMC (g)	BMD (g/cm ²)	T - score	PR (%)	Z - score	AM (%)
L1	14.41	14.44	1.002	0.7	108	2.0	129
L2	15.27	16.33	1.069	0.4	104	1.8	123
L3	16.99	19.69	1.159	0.7	107	2.2	127
L4	18.74	21.27	1.135	0.2	102	1.8	121
Total	65.41	71.72	1.096	0.4	105	1.9	124

Total BMD CV 1.0%, ACF = 1.000, BCF = 1.000, TH = 3.855

WHO Classification: Normal
Fracture Risk: Not Increased



Physician's Comment:

HOLOGIC®

Osteoporosis treatment

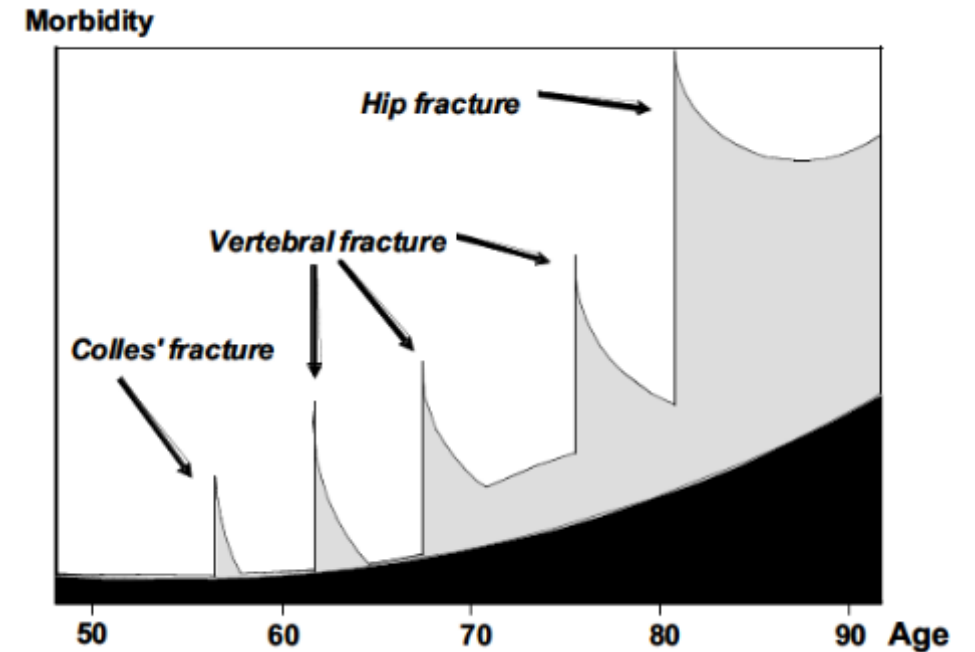


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- Non pharmacological
 - Exercise
 - Sunlight
 - Nutritional (Vitamin D + Calcium)
 - Risk factor modification
- Pharmacological
 - Bisphosphonates (Alendronate)
 - Teriparatide (PTH analogue)
 - Denosumab (RANKL inhibitor)
 - Strontium ranelate
 - Selective Estrogen Receptor Agonists (SERMs) (Raloxifene)
 - Hormone replacement therapy

Fracture and Quality of life over the life span

- ...seriously damage the quality of life of survivors.
- >50% cannot return to their premorbid state.
- 10% needs to find extra help in daily care.



Adapted from *J Endo Investigation* 1999;22(8):583-588 Kanis JA & Johnell O

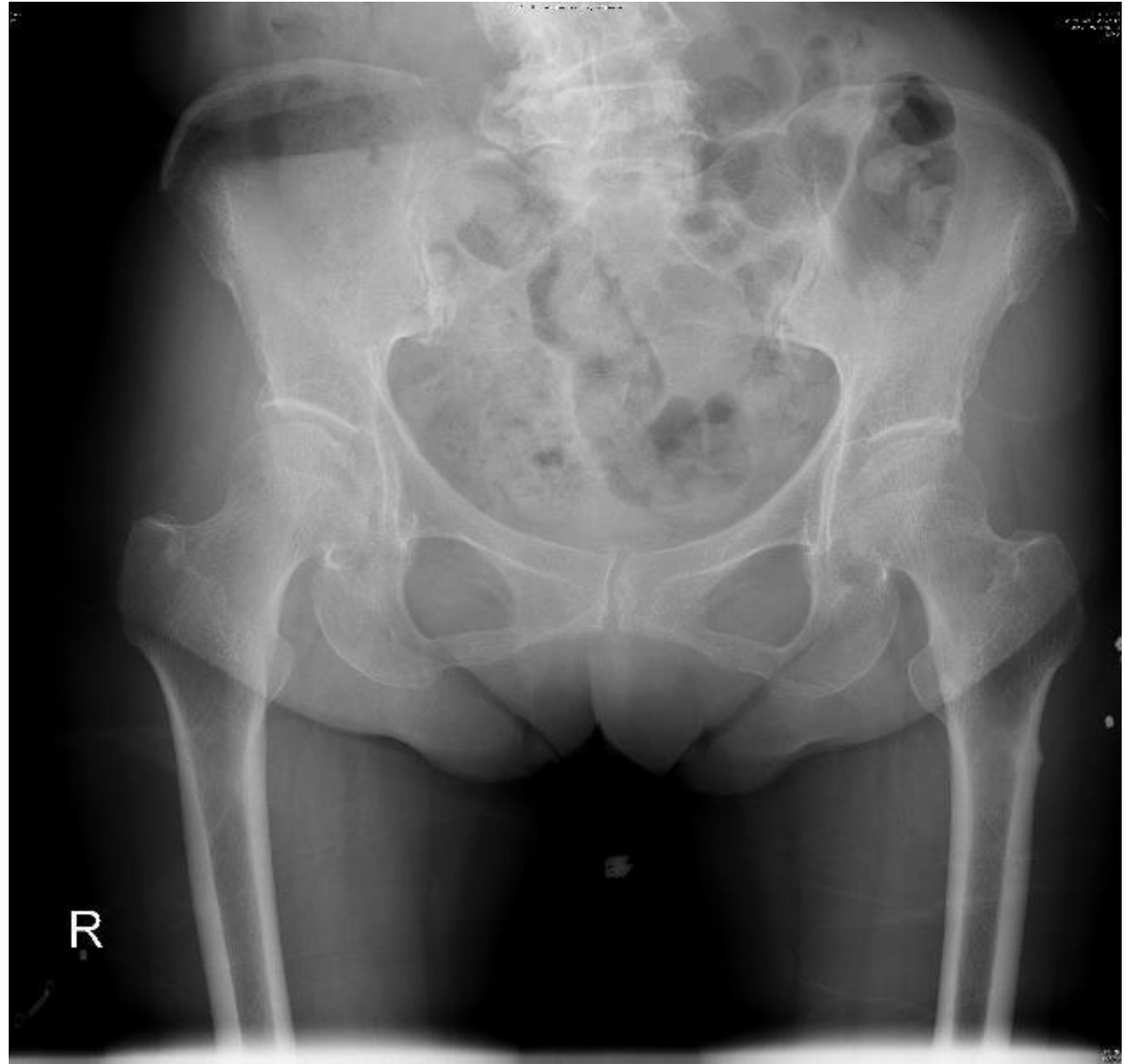
F 72 – Left thigh pain – 6 years bisphosphonates



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Mechanical pain with walking

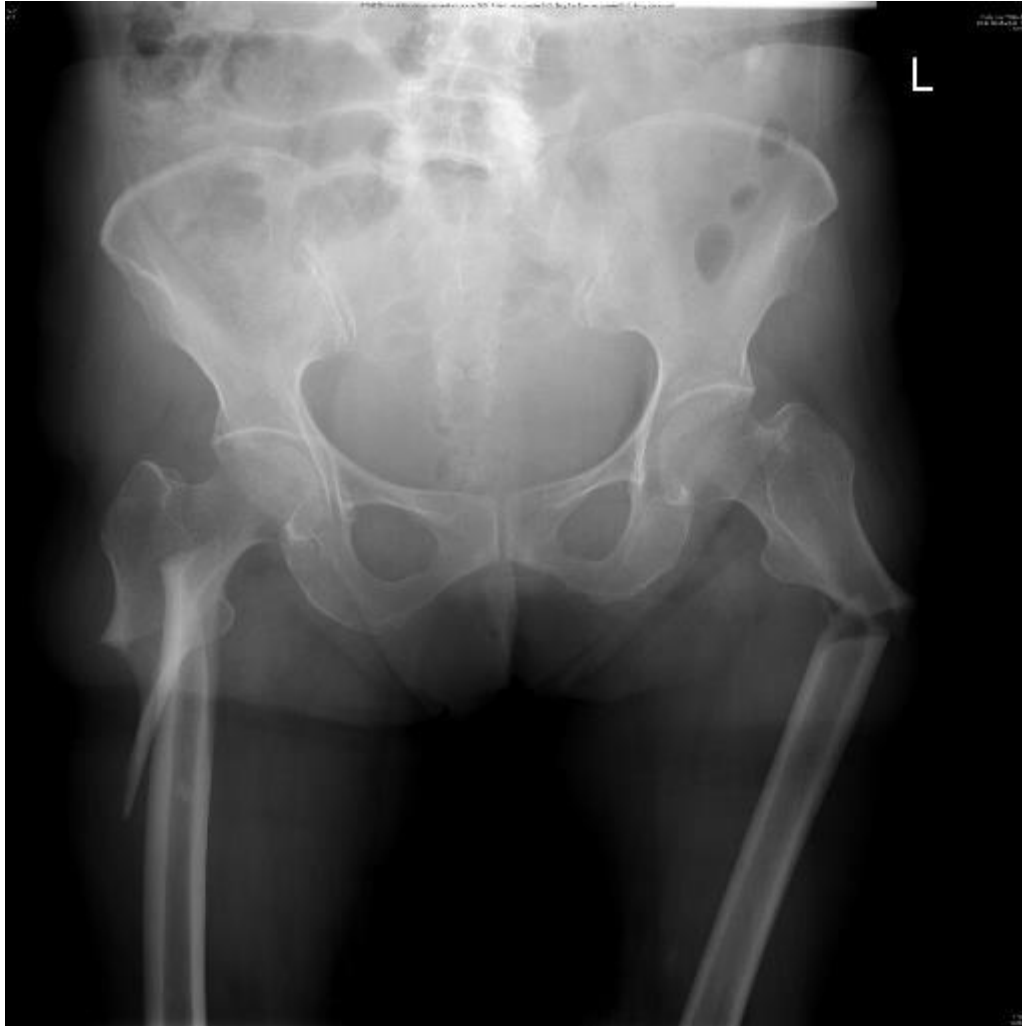
Percussion, rotation and axial loading



F 72 – tripped in toilet – prodromal bilateral pain for 5 months

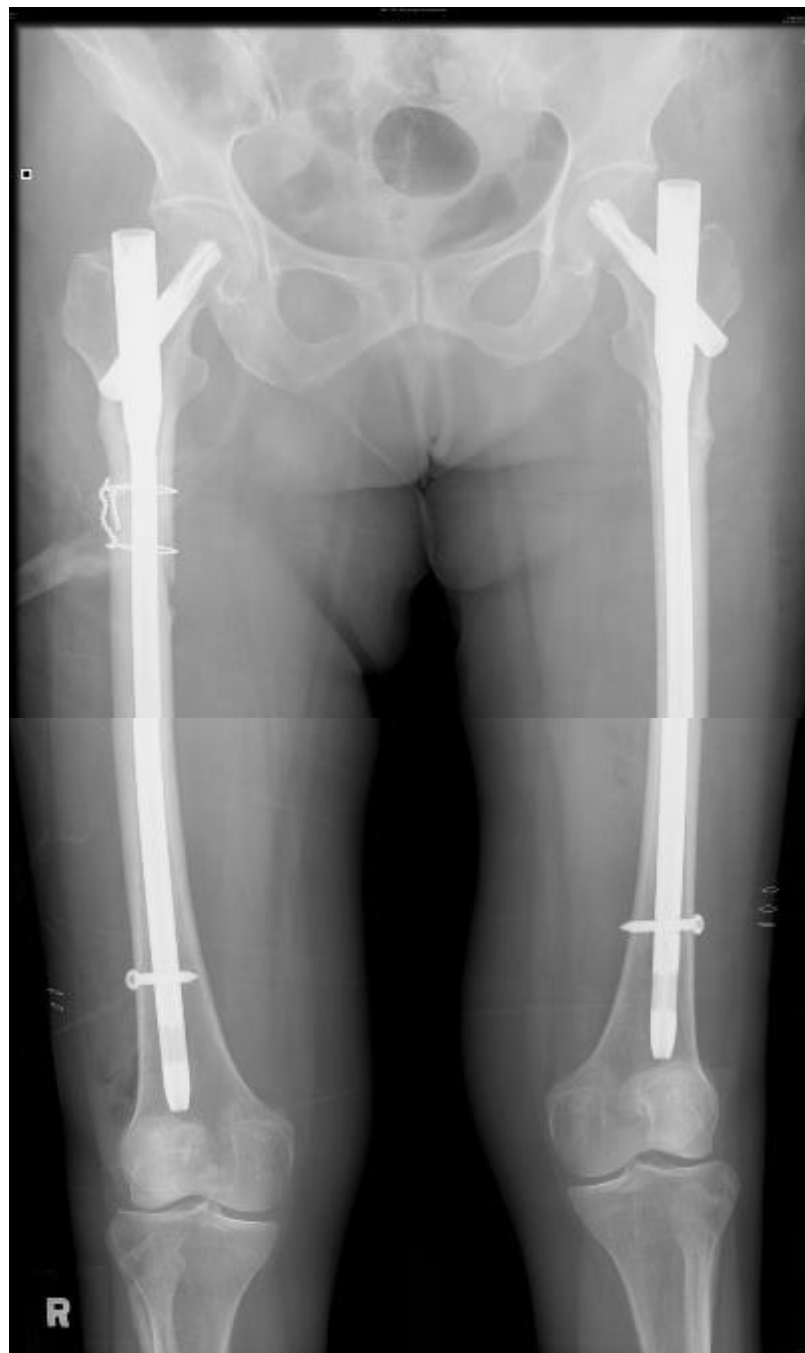


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Beak sign

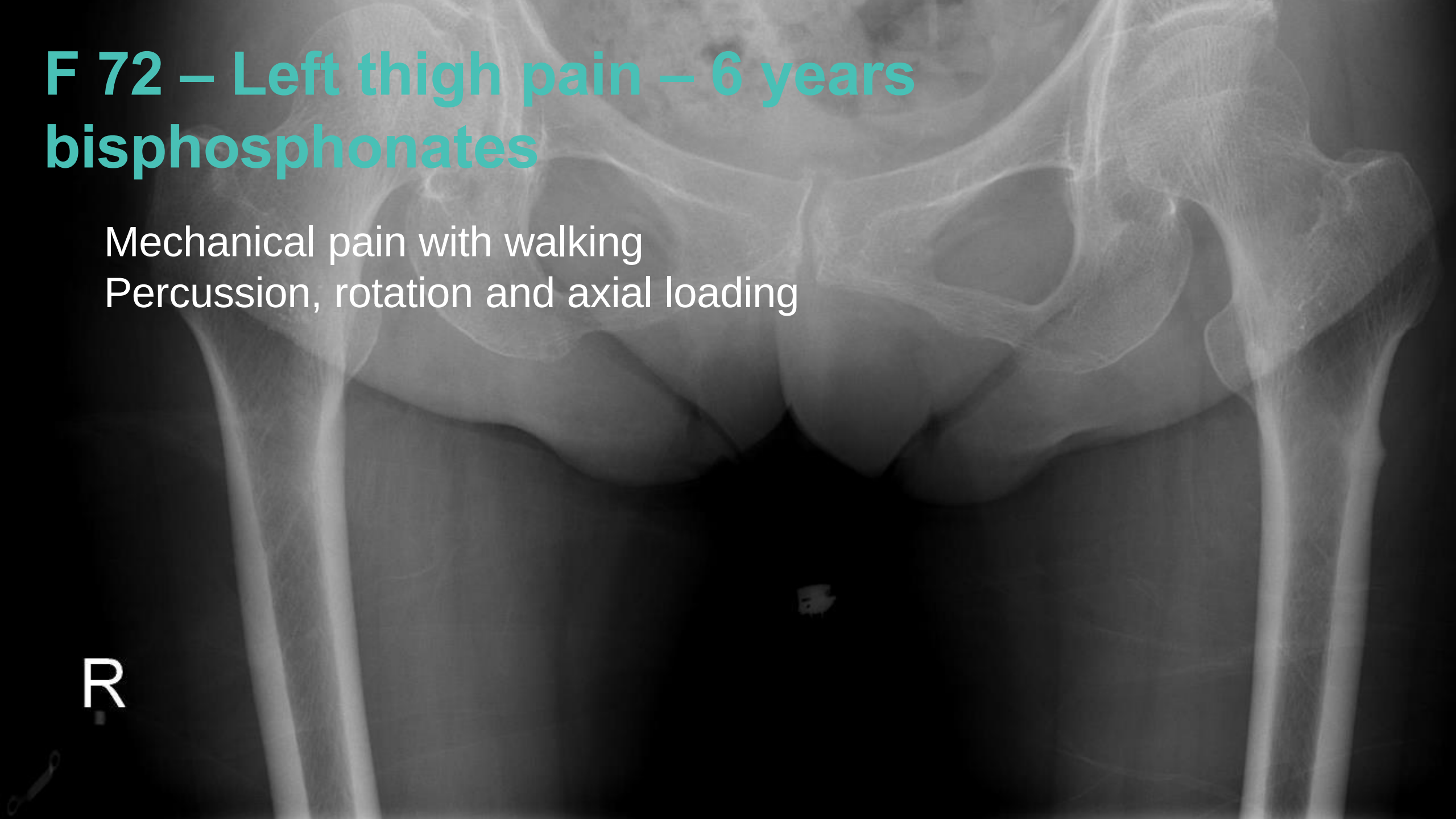


Boths sided Open reduction

F 72 – Left thigh pain – 6 years bisphosphonates

Mechanical pain with walking
Percussion, rotation and axial loading

R

An anteroposterior (AP) radiograph of the pelvis and proximal femurs. The image shows the bony structures of the pelvis, including the iliac, ischial, and pubic bones, as well as the proximal femurs. The left femur (on the right side of the image) shows a subtle, ill-defined area of increased radiopacity in the proximal shaft, which is the area of concern for the patient's pain. The right femur (on the left side of the image) appears normal. The label 'R' is visible in the bottom left corner, indicating the right side of the patient.

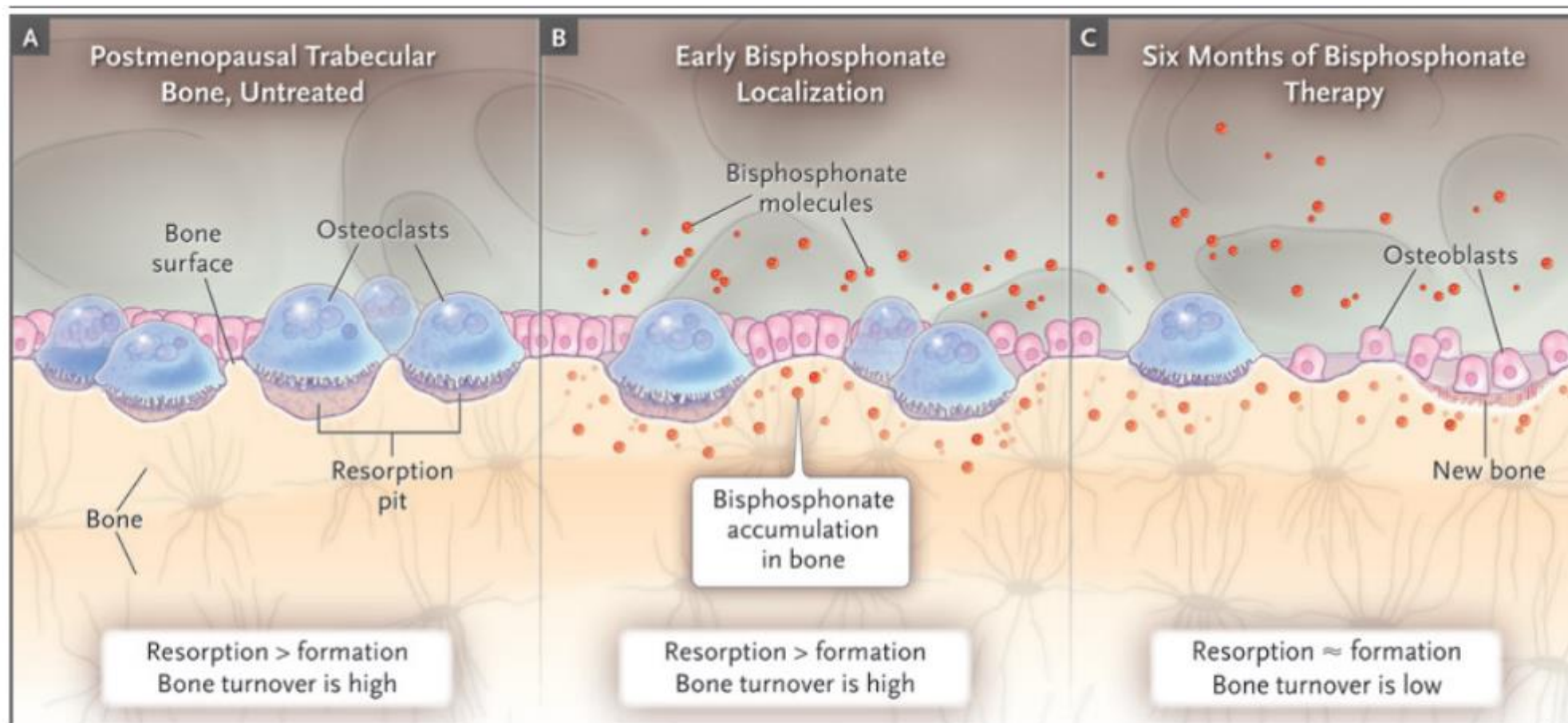
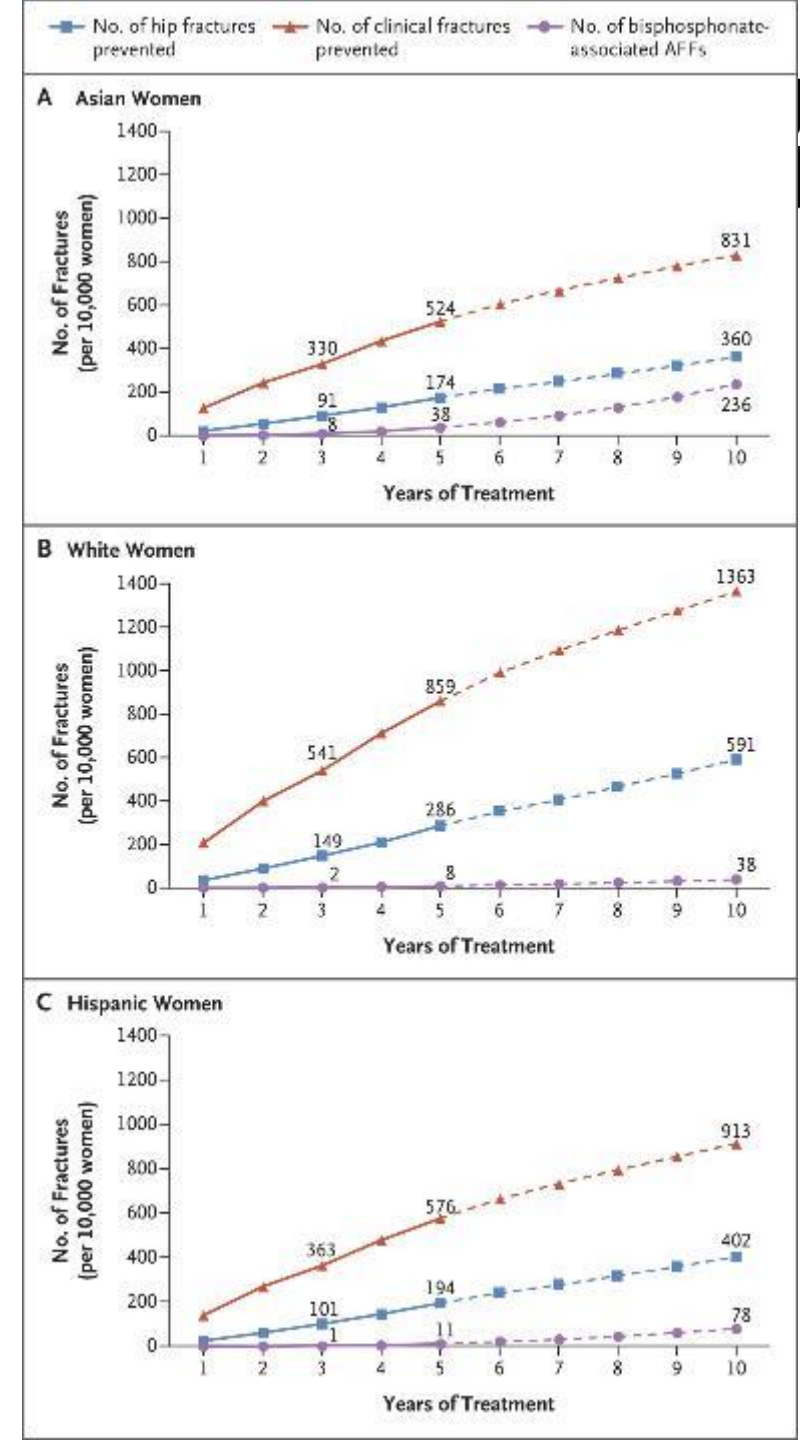


Figure 3. Cellular Elements Involved in Postmenopausal Trabecular Bone Turnover before and during Bisphosphonate Therapy.

Panel A shows the untreated postmenopausal state, in which osteoclast-mediated bone resorption occurs at a high rate, exceeding osteoblast-driven bone formation and leading to net bone loss. Panel B shows the initial events associated with bisphosphonate therapy, including the localization and concentration of bisphosphonates in bone through binding to sites of active bone resorption. Panel C shows the effects of bisphosphonates after 6 months of therapy. The number of osteoclasts has decreased owing to early apoptosis. As a result, bone resorption is decreased, and osteoblasts and bone formation are also decreased. The bisphosphonate concentration is reduced around previous resorption pits. A lower steady-state rate of bone turnover, similar to premenopausal rates, is established.

Atypical femur fractures

- Reduced bone turnover
- Microfractures accumulation
- Stress fracture
- Eventual displacement with minimal trauma
- ‘Beak’ sign on Xray and ‘dreaded black line’
- Relation with bisphosphonate use > 3 years
- Benefit of bisphosphonates still > risks





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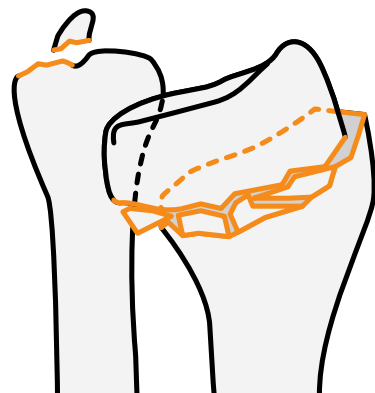
Distal radial fractures—the problem



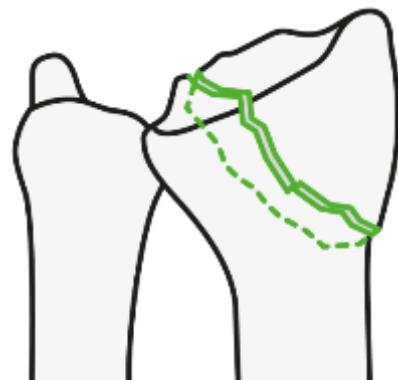
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Common





A: Extraarticular
(within width of joint) –
*Colles or Smith Fracture



B: “Partial” articular –
*Barton fracture



C: “Complete” articular

Changes in wrist mechanics may lead to arthrosis especially with intraarticular steps



Problems with acute displacement



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- Pain
- Unstable
- Nerve impingement
- Loss of reduction
- Skin and tendon impingement

Principle of fracture management?



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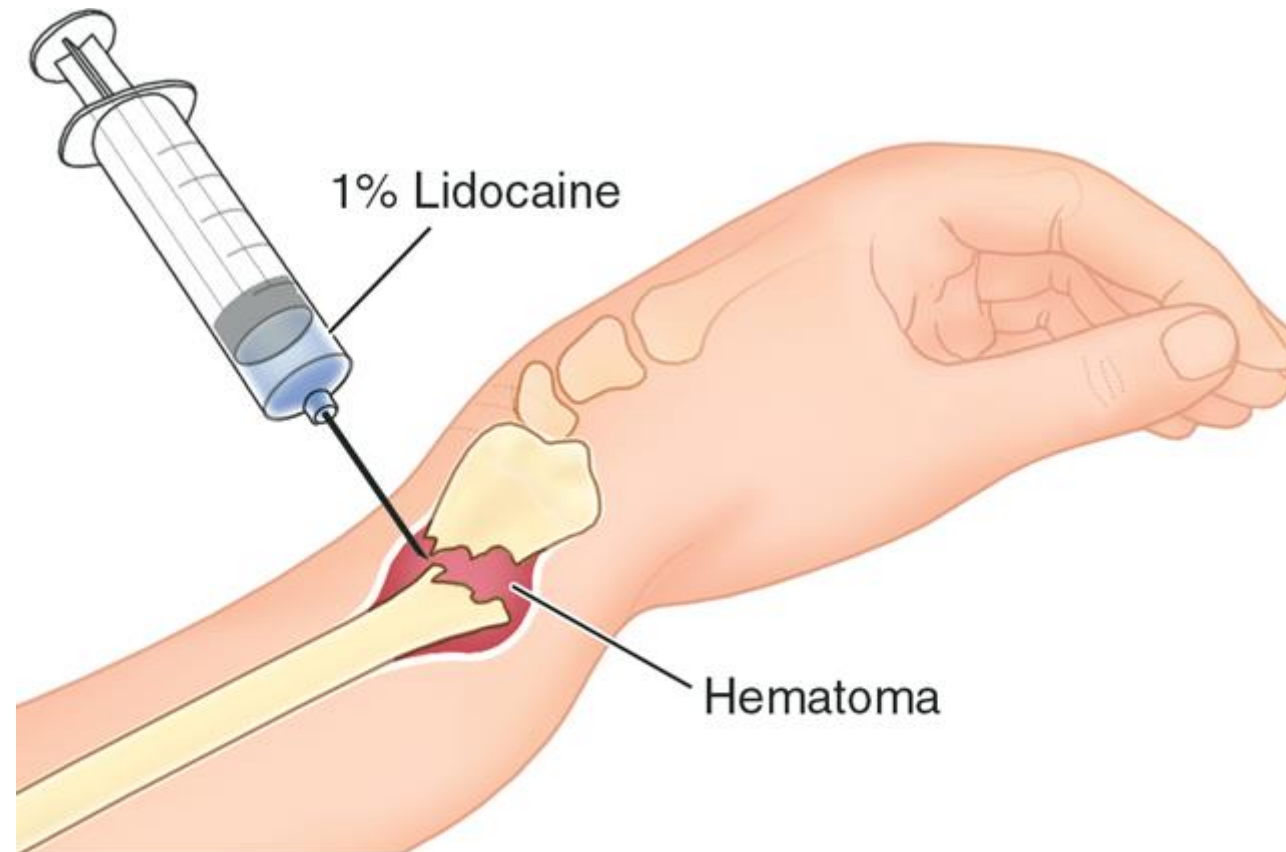
- Reduction if necessary
- Stabilization if necessary
- Rehabilitation always



Hematoma block



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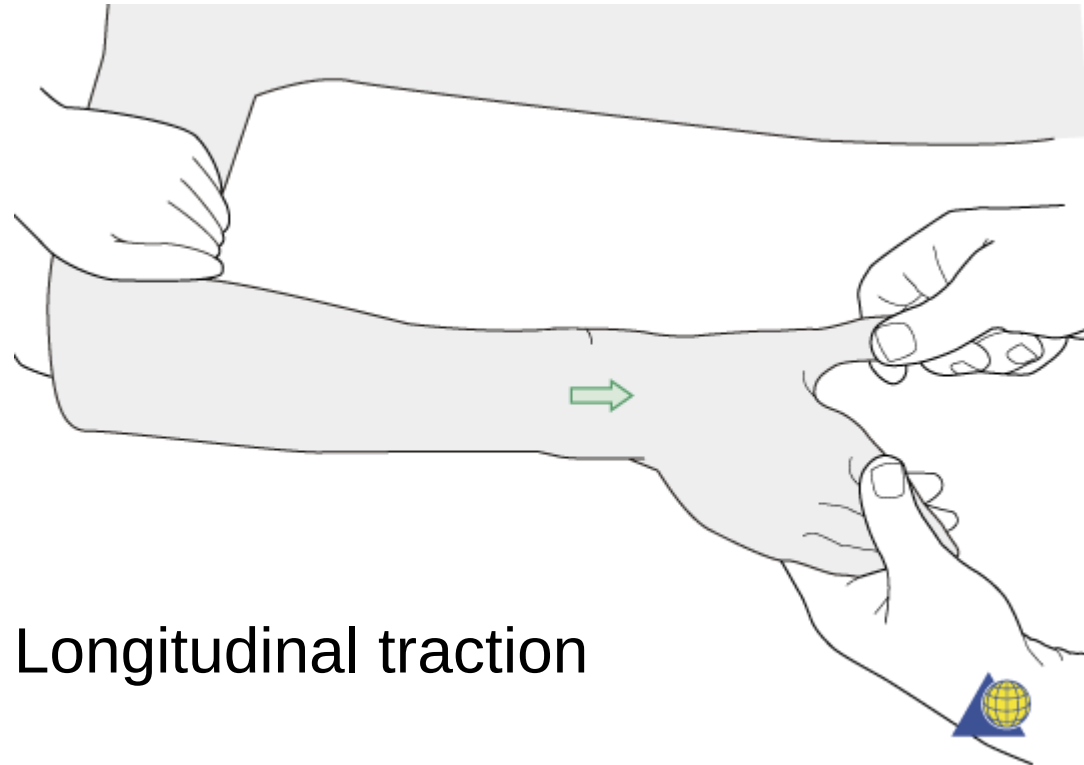


Source: Reichman EF: *Emergency Medicine Procedures*,
Second Edition: www.accessemergencymedicine.com
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Closed reduction



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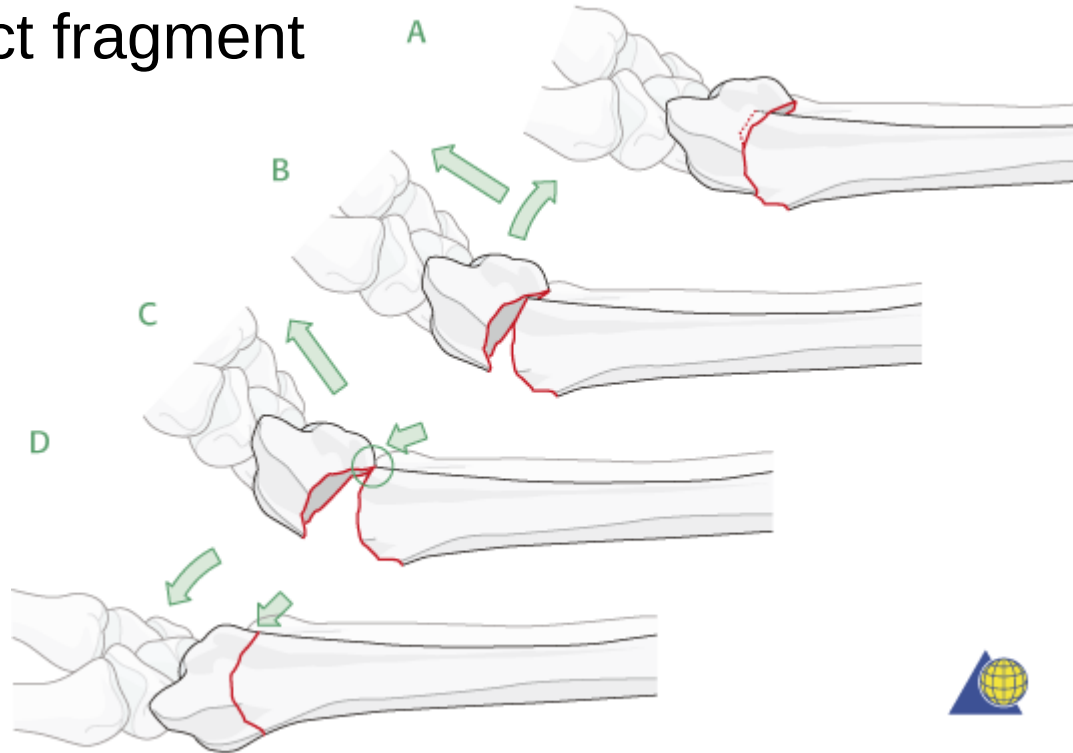


Longitudinal traction

Closed reduction



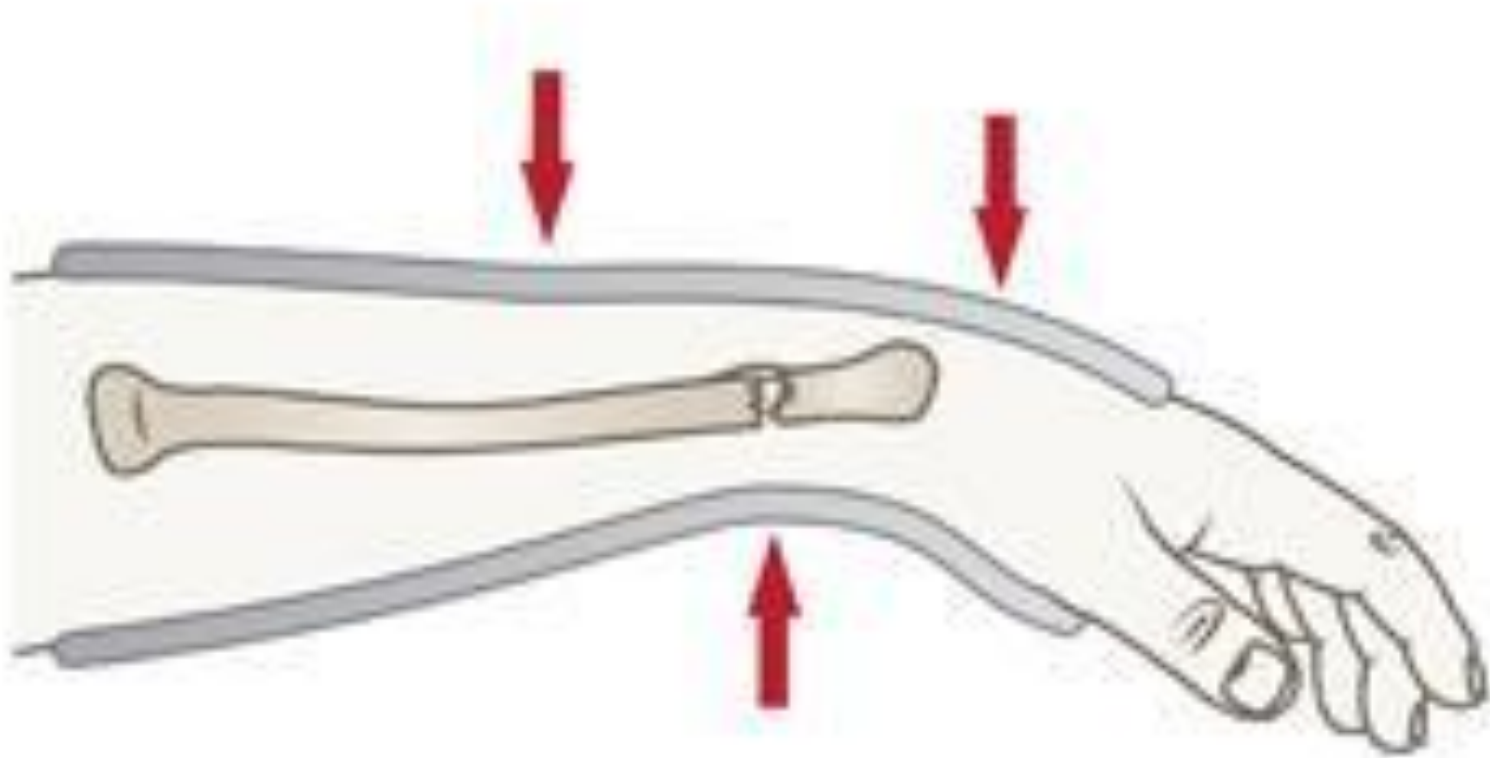
Disimpact fragment



Casting after Closed reduction



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3-point fixation

Closed reduction and cast—BEWARE



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- Bad cast – too tight or too loose
- Loss of reduction



Operative treatment of distal radial fractures—decision making



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- Evaluate patient's functional demand
- Do a neurological exam (median nerve)
- Fluoroscopy
- CT assessment



Who needs surgery?



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- **Most do NOT need an operation**
 - Non-displaced fractures
 - Well reduced stable fractures
 - Very low demand patients
 - Geriatric

Absolute indications for surgery



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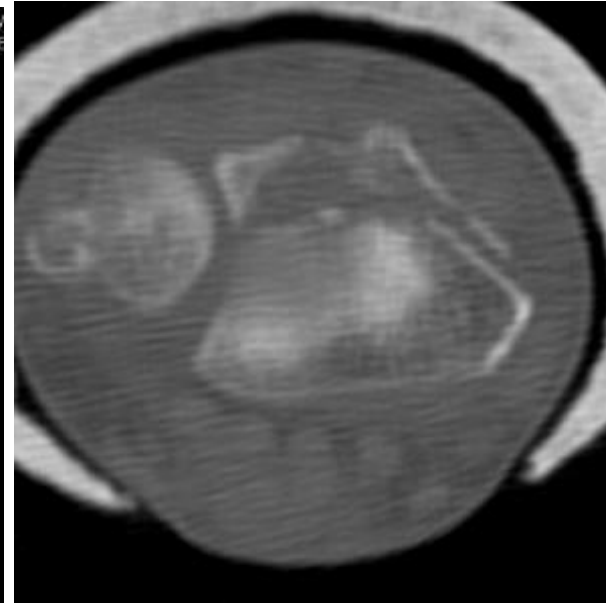
- High grade open fracture
- Acute (progressing) median nerve palsy
- Associated carpal dislocation



Open reduction Internal Fixation with Locking plate



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Intraarticular
Displaced #



Progress after treatment



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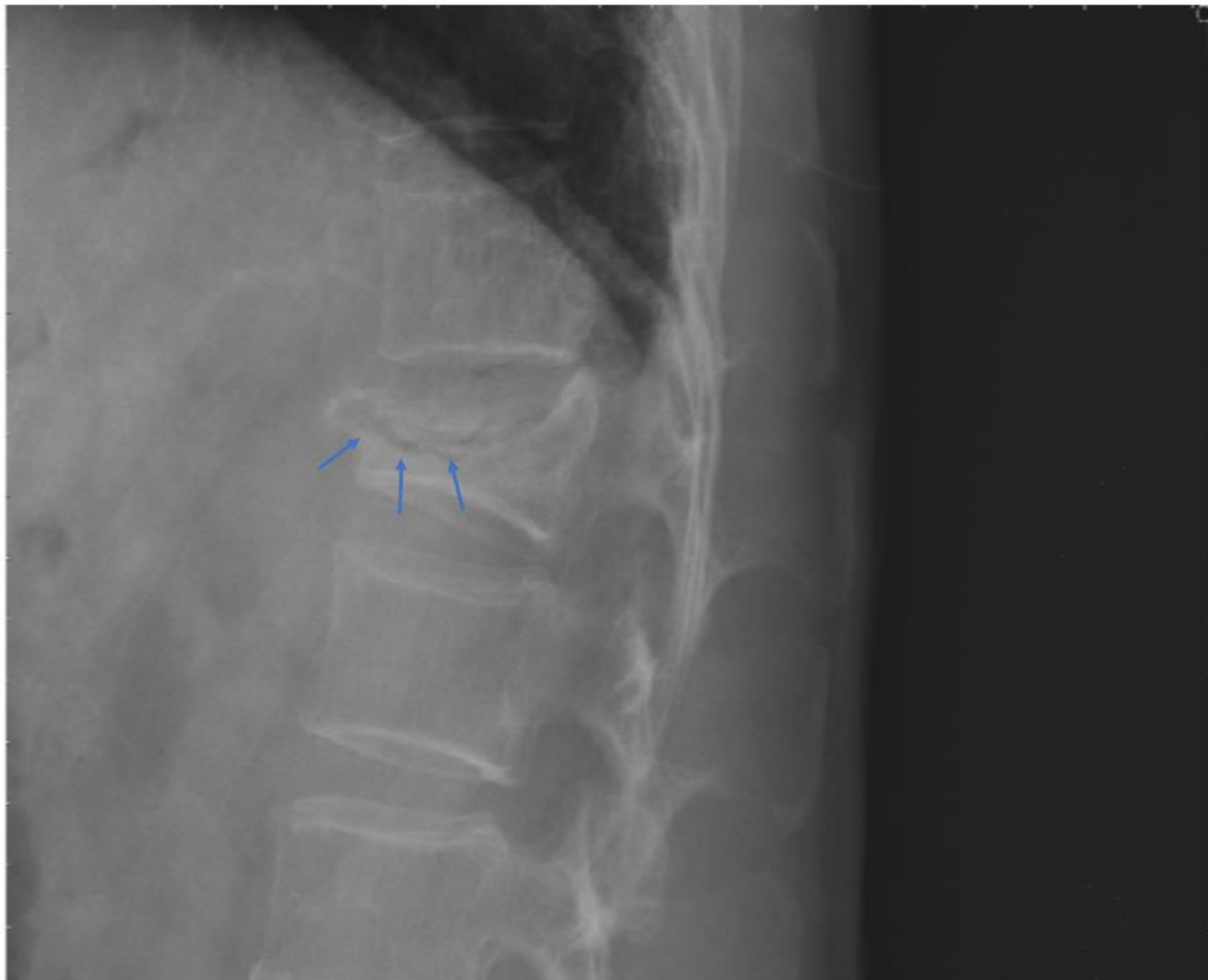
- Cast immobilization
 - Bone Healing after 4 weeks
 - Physiotherapy / Occupational therapy
 - Range of motion exercise and strengthening
- Surgical repair
 - Immediate stability
 - Early mobilization
 - Strengthening after 4-6 weeks

F70, Fall injury and back pain

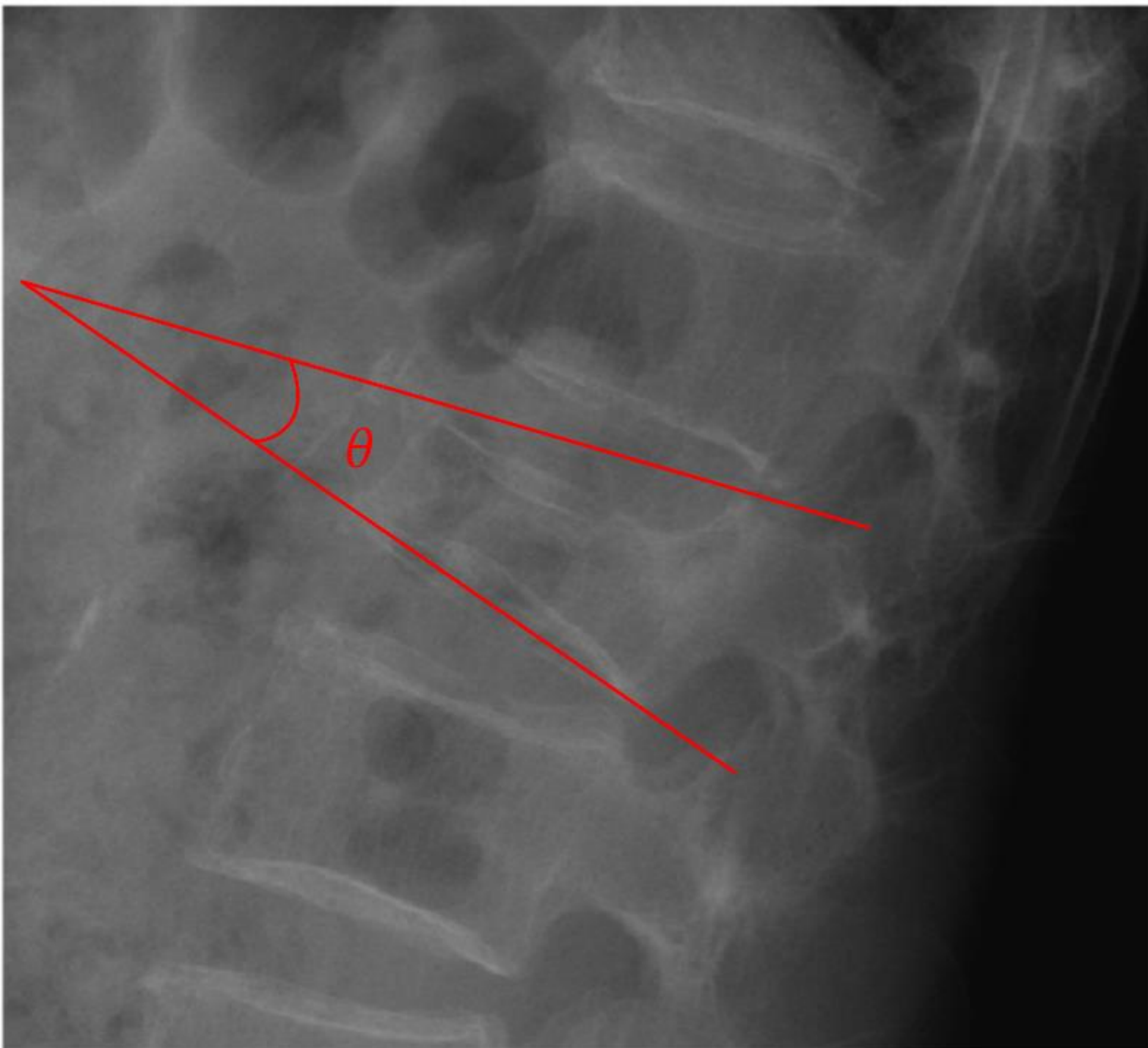


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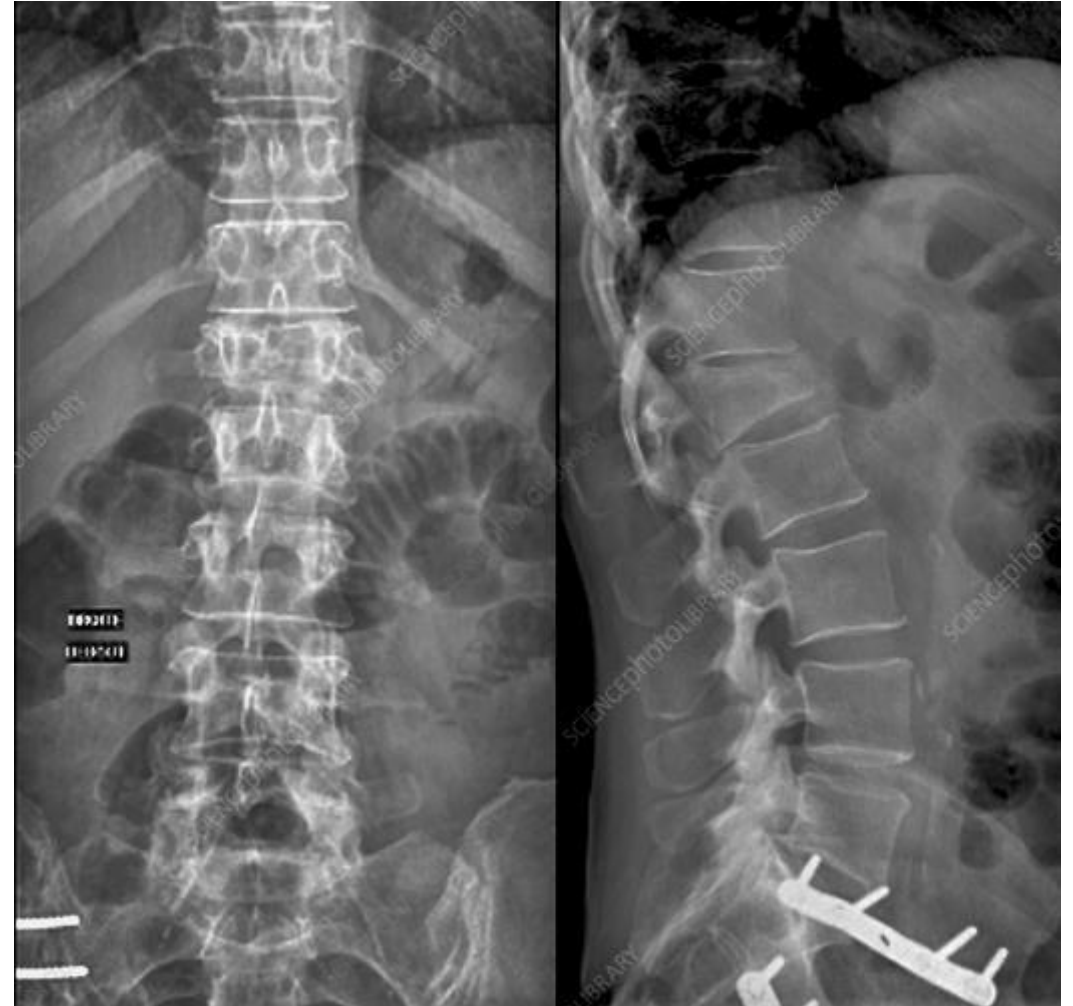
Thoracic
kyphosis

Principle of fracture management?



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- Reduction if necessary
- Stabilization if necessary
- Rehabilitation always

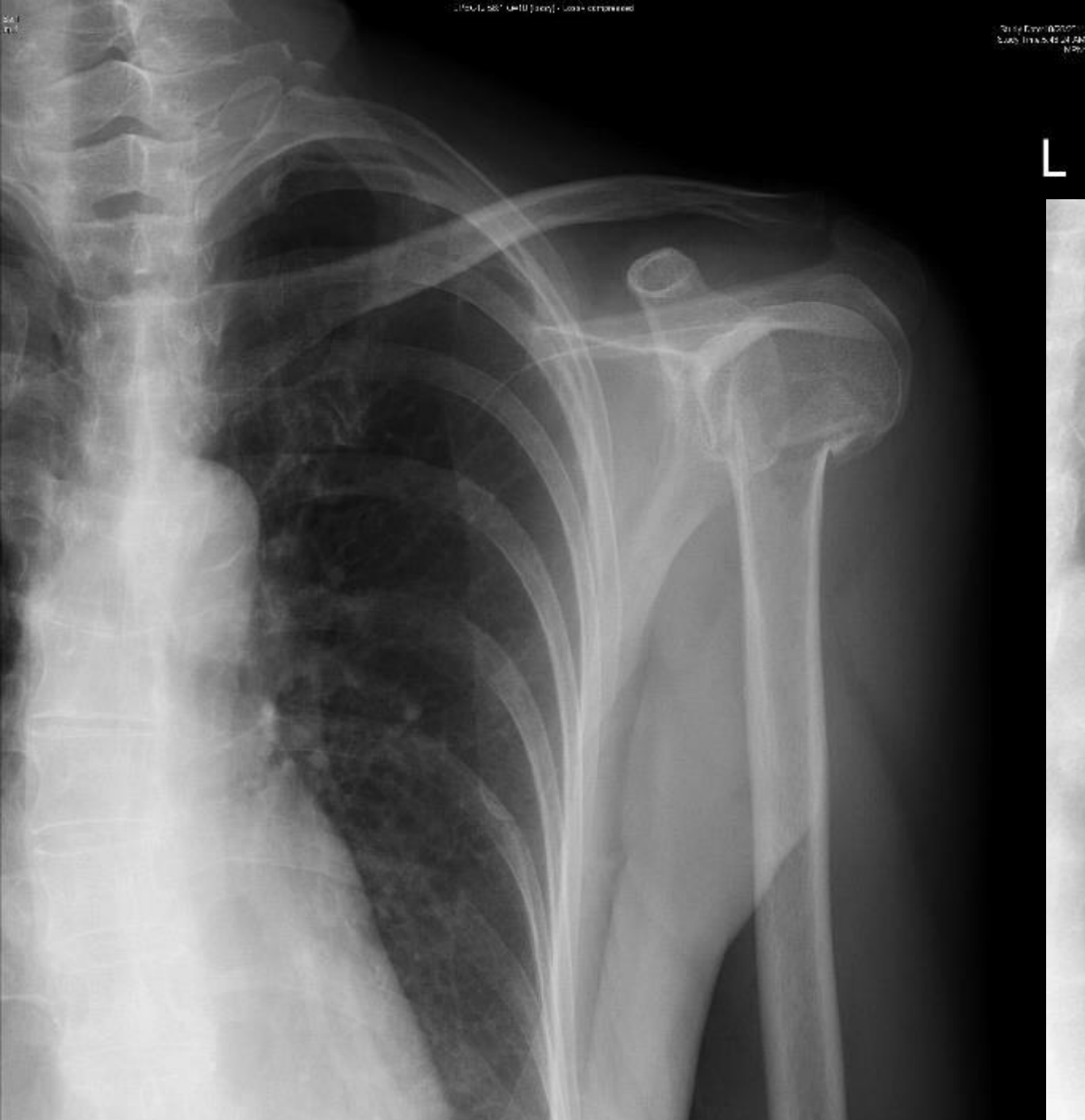


Other common osteoporotic injuries



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- Vertebral collapse
- Proximal humerus
- Olecranon
- Distal femur
- Pelvic insufficiency fracture



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(R)

R



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Upper vs Lower limb



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- Lower limb – for locomotion
 - Difficulty pain control with weight bearing
 - Significant morbidity with displacement / delayed union
- Upper limb – for self care function
 - Better managed by immobilization
 - Usually recovers with time and rehabilitative training

Suspected hip fracture – normal X-ray



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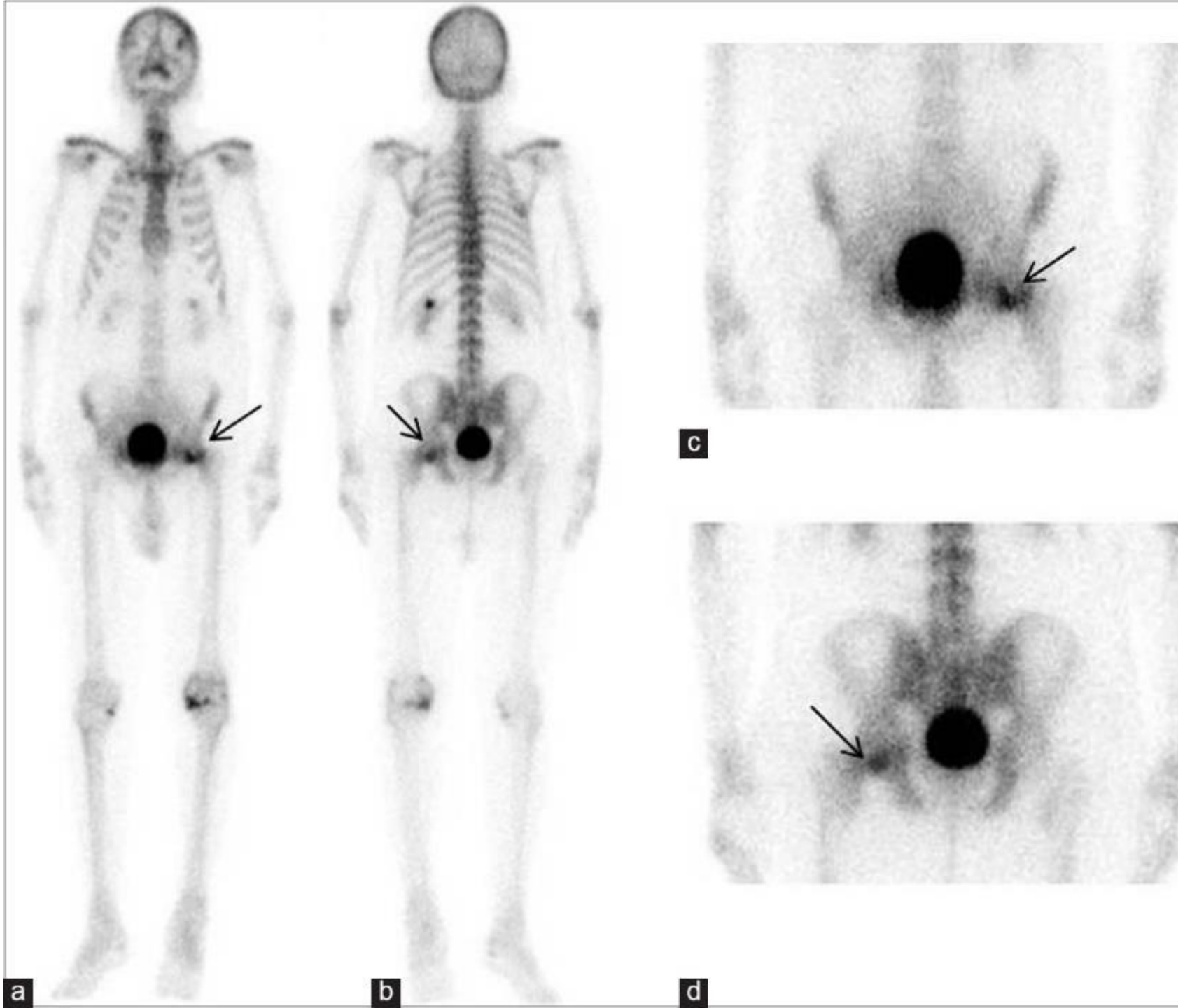
- Old age patient with fall and hip pain
- Unable to walk
- Pain on weight bearing

- Repeated X-rays – no abnormality
- Use advanced imaging
 - MRI (Most sensitive)
 - Bone Scan
 - CT



Bone edema (Water) on MRI T2





General Assessment



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- Mode of injury
- Associated injury
- General state
- Medical conditions
- Local features

Management



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- Avoid extensive immobilization
- Pain control
- Early function
- Mx of Osteoporosis
- Fall prevention

Hip fractures



- Intracapsular (neck of femur)
 - Young age vs Old age
 - Displaced vs undisplaced
- Extracapsular (Trochanteric)
 - Stable vs Unstable
- Shaft fracture extension
- Atypical femur fractures (bisphosphonate related)

Distal Radius fractures



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- Colles fracture
- Smith fracture
- Barton fracture



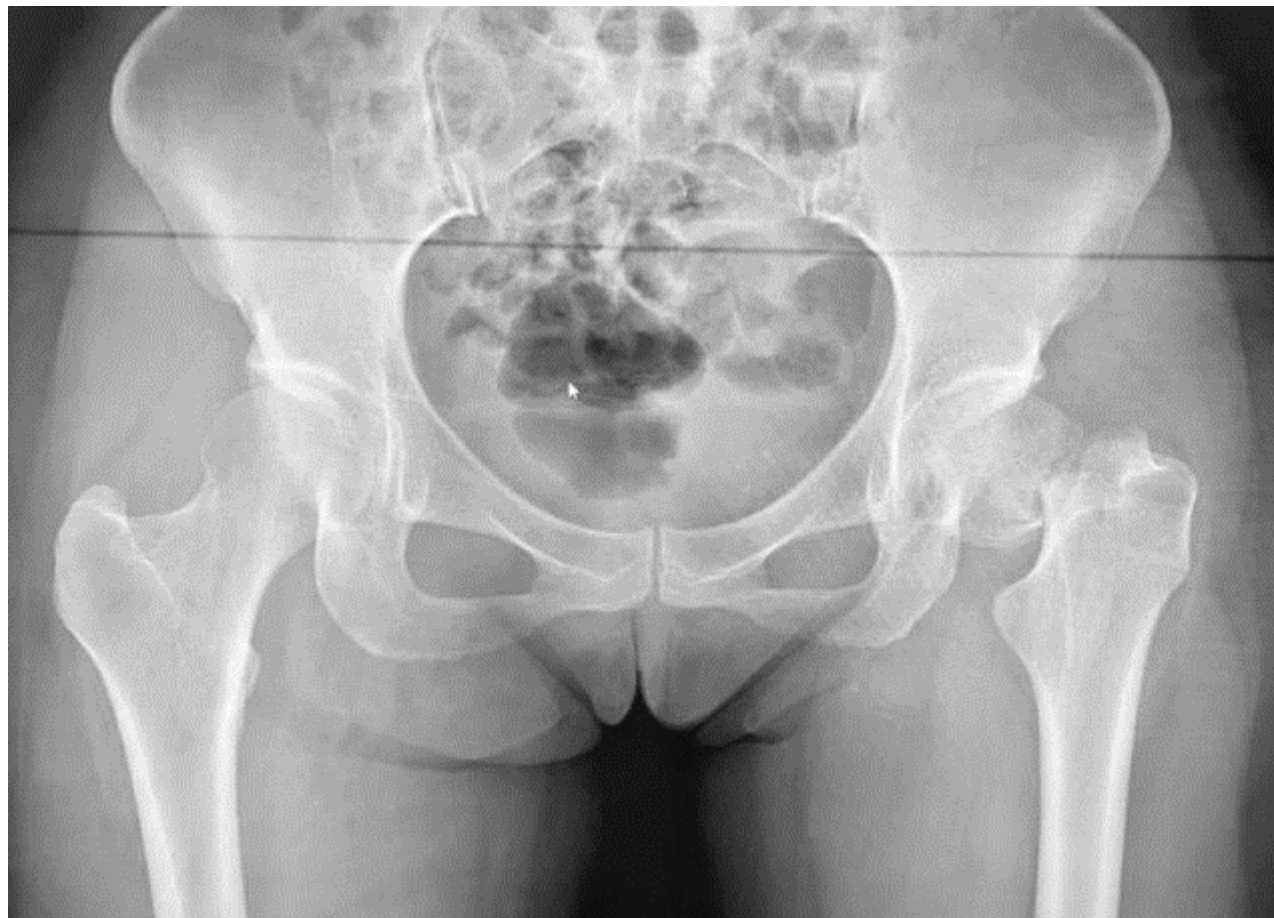
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Practise !

41/F Garden IV



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Immediate post op

Total Tip apex index = 12mm



6 months

- Union achieved
- No collapse/ shortening of fracture site
- Returned to Recreational walking status

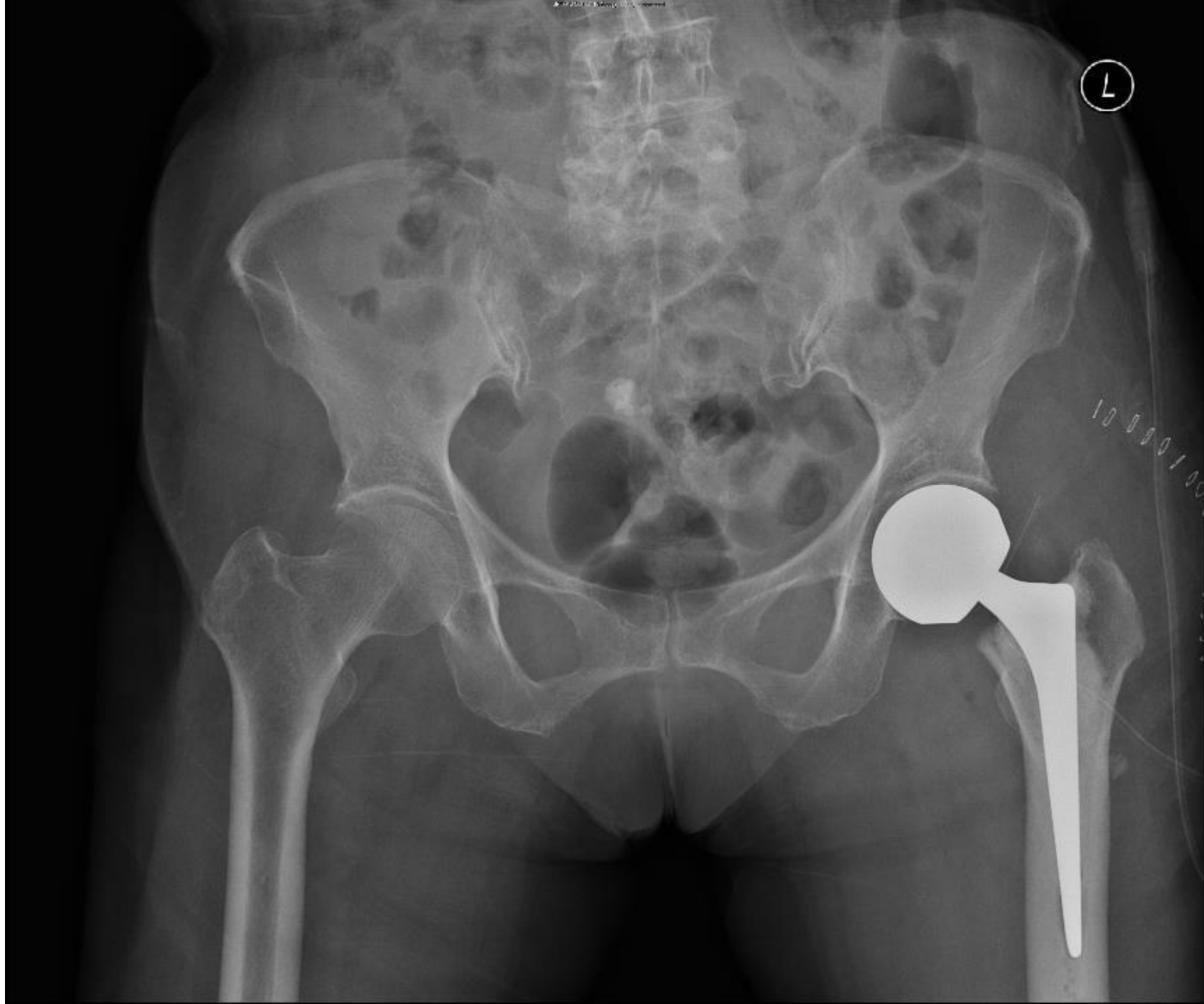




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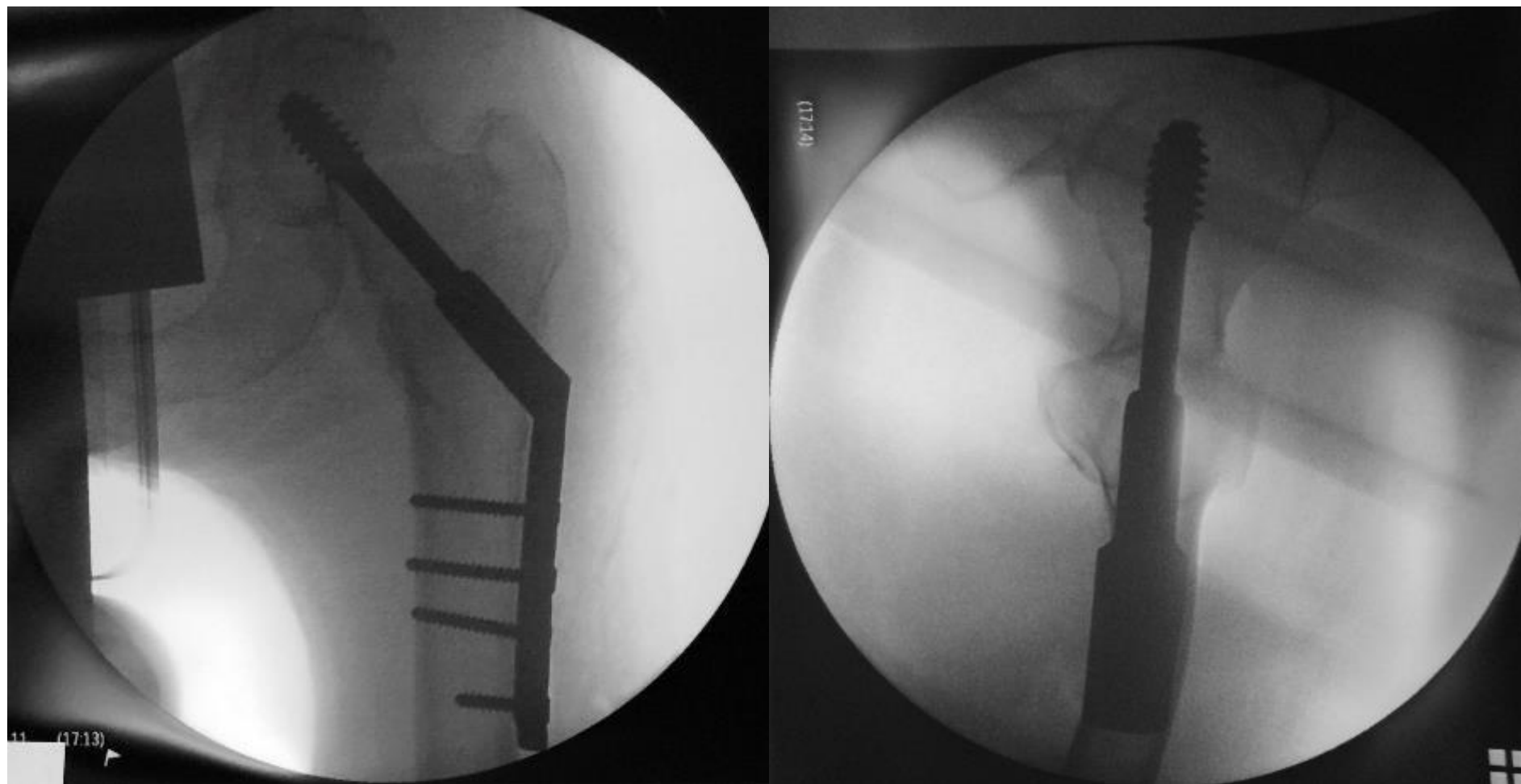
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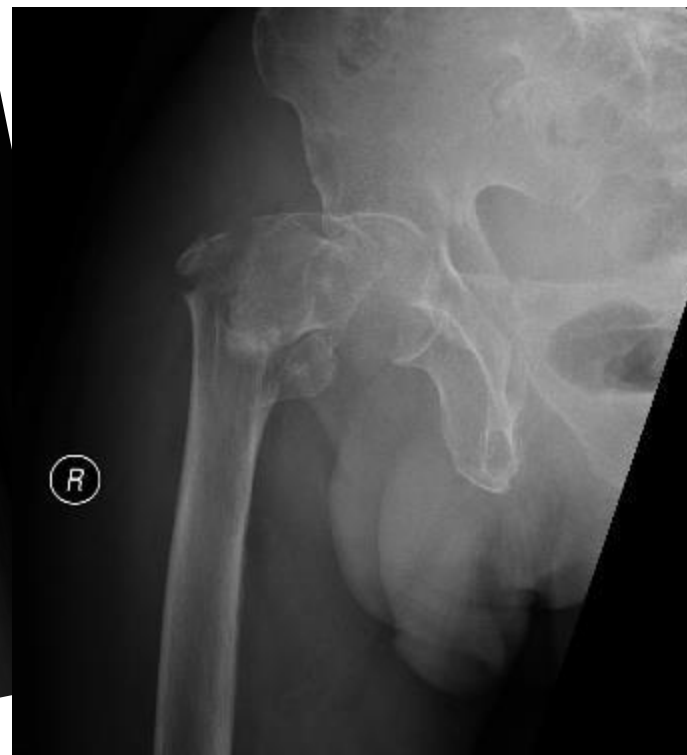








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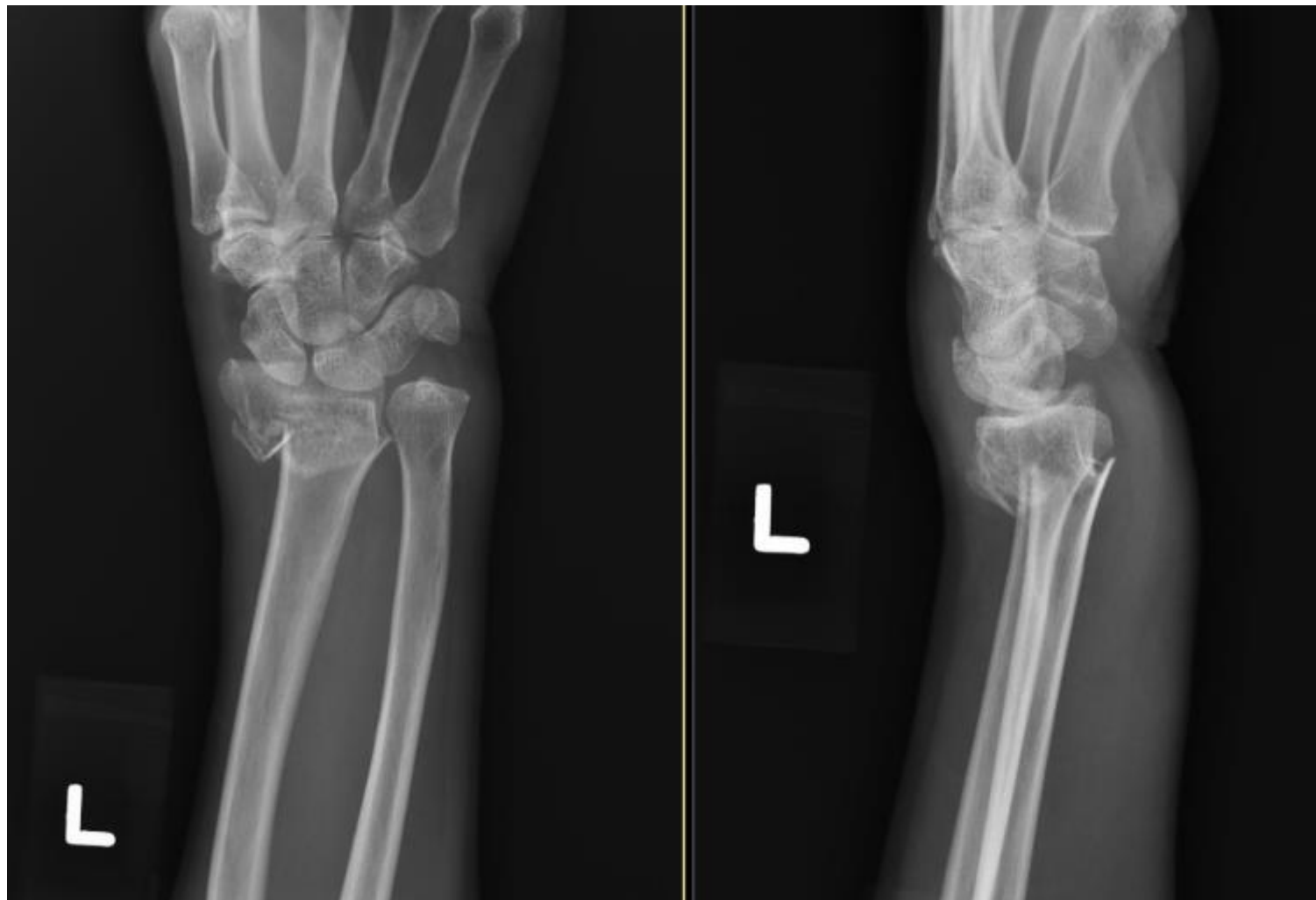




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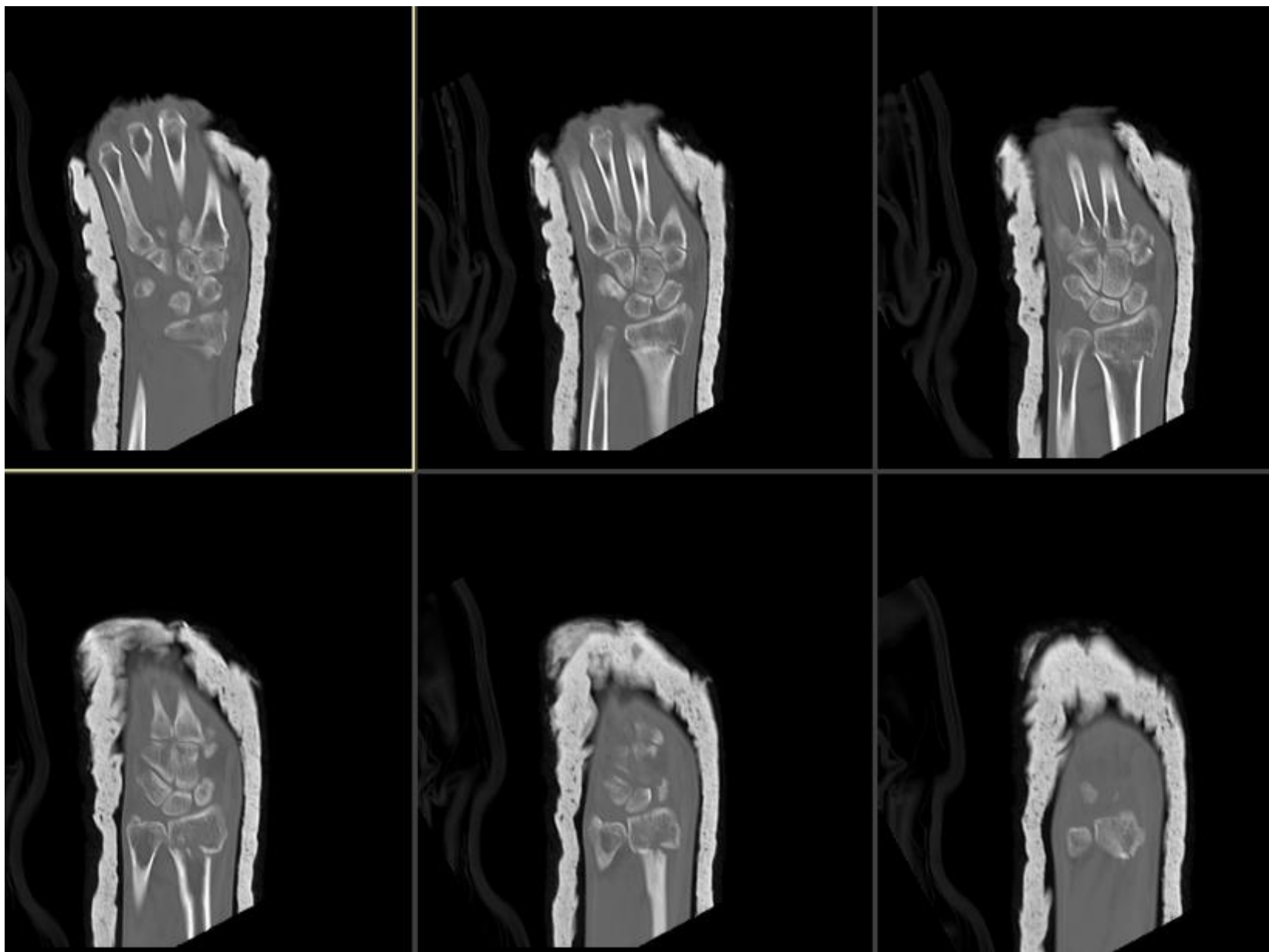


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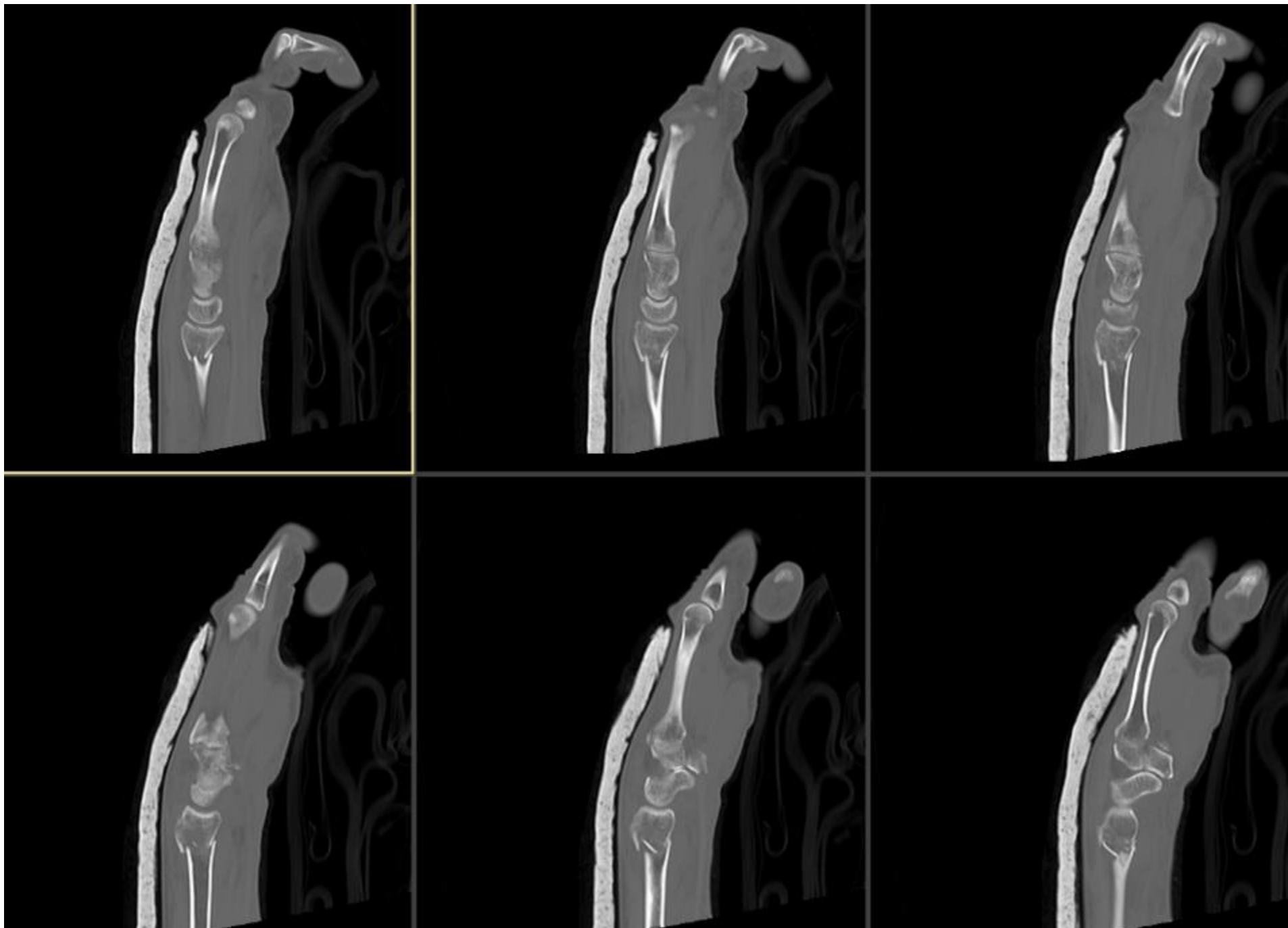


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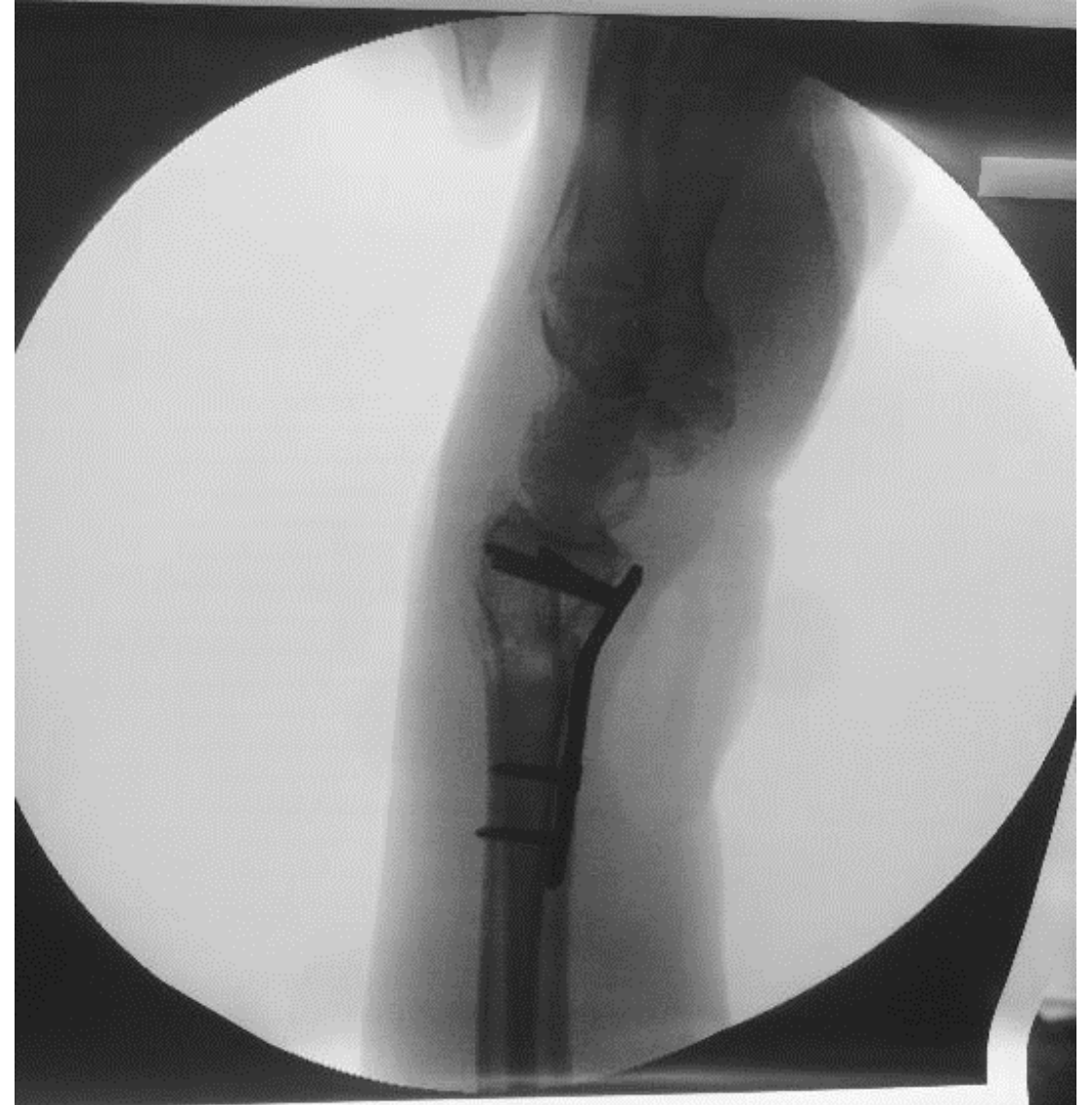
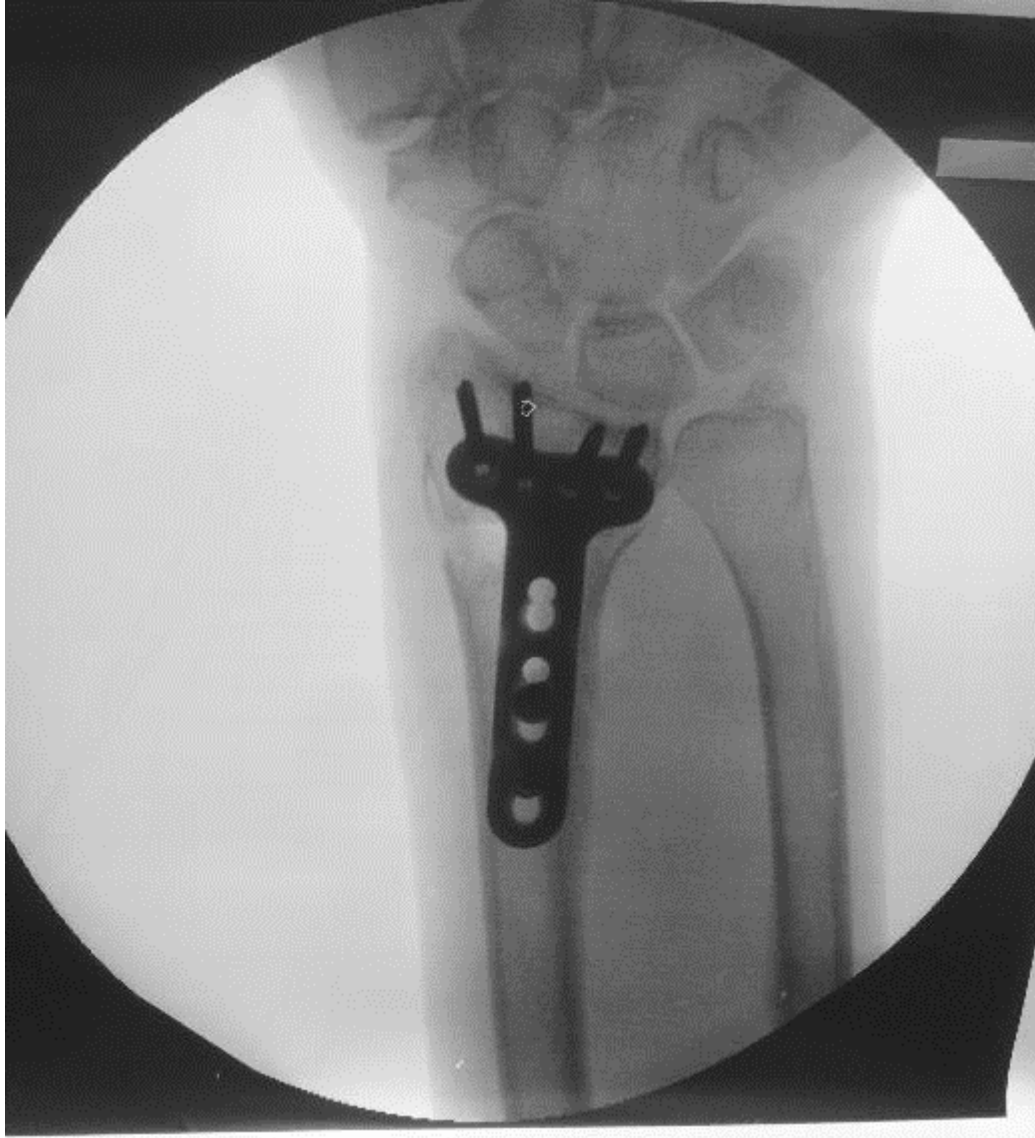
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Open reduction internal fixation



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